HBSC

Updated: 2026-09-06

Data Overview

It's a huge file:

```
## [1] 244097    120
```

With these columns:

HBSC	seqno_int	cluster	countryno	region
id1	id2	id3	id4	weight
adm	month	year	age	agecat
sex	grade	monthbirth	yearbirth	fasfamcar
fasbedroom	fascomputers	fasbathroom	fasdishwash	fasholidays
health	lifesat	headache	stomachache	backache
feellow	irritable	nervous	sleepdificulty	dizzy
thinkbody	physact60	breakfastwd	breakfastwe	fruits_2
vegetables_2	sweets_2	softdrinks_2	fmeal	toothbr
timeexe	smokltm	smok30d_2	alcltm	alc30d_2
drunkltm	drunk30d	cannabisltm_2	cannabis30d_2	bodyweight
bodyheight	likeschool	schoolpressure	studtogether	studhelpful
studaccept	teacheraccept	teachercare	teachertrust	bulliedothers
beenbullied	cbulliedothers	cbeenbullied	fight12m	injured12m
friendhelp	friendcounton	friendshare	friendtalk	emconlfreq1
emconlfreq2	emconlfreq3	emconlfreq4	emconlpref1	emconlpref2
emconlpref3	emcsocmed1	emcsocmed2	emcsocmed3	emcsocmed4
emcsocmed5	emcsocmed6	emcsocmed7	emcsocmed8	emcsocmed9
hadsex	agesex	contraceptcondom	contraceptpill	countryborn
countrybornmo	countrybornfa	motherhome1	fatherhome1	stepmohome1
stepfahome1	fosterhome1	elsehome1_2	employfa	employmo
employnotfa	employnotmo	talkfather	talkstepfa	talkmother
talkstepmo	famhelp	famsup	famtalk	famdec
MBMI	IRFAS	IRRELFAS_LMH	IOTF4	oweight_who

So after selecting and grouping variables, create dat with :

```
## ## Individual Information
## * `seqno_int`
## * `countryno`
## * `agecat`
## * `sex`
## * `MBMI`
## * `IRFAS`
```

```
## * `IRRELFAS_LMH`
## * `IOTF4`
## * `bodyweight`
## * `bodyheight`
##
## ## Health Issues and Life Satisfaction
## * `health`
## * `lifesat`
## * `headache`
## * `stomachache`
## * `backache`
## * `feellow`
## * `irritable`
## * `nervous`
## * `sleepdificulty`
## * `dizzy`
##
## ## Problematic Social Media Use (PMSU)
## * `emcsocmed1`
## * `emcsocmed2`
## * `emcsocmed3`
## * `emcsocmed4`
## * `emcsocmed5`
## * `emcsocmed6`
## * `emcsocmed7`
## * `emcsocmed8`
## * `emcsocmed9`
##
## ## Bullying
## * `beenbullied`
## * `cbeenbullied`
##
## ## Communicates Online More Than Real Life
## * `emconlfreq1`
## * `emconlfreq2`
## * `emconlfreq3`
## * `emconlfreq4`
## * `emconlpref1`
## * `emconlpref2`
## * `emconlpref3`
##
## ## Family Support
## * `famhelp`
## * `famsup`
## * `famtalk`
## * `famdec`
##
## ## Friends Support
## * `friendhelp`
## * `friendcounton`
## * `friendshare`
## * `friendtalk`
##
## ## School Support
```

```
## * ‘likeschool‘
## * ‘schoolpressure‘
## * ‘studtogether‘
## * ‘studhelpful‘
## * ‘studaccept‘
## * ‘teacheraccept‘
## * ‘teachercare‘
## * ‘teachertrust‘
##
## ## Communication with Parents
## * ‘talkfather‘
## * ‘talkmother‘
##
## ## Good/Bad Habits
## * ‘physact60‘
## * ‘timeexe‘
## * ‘alc30d_2‘
```

The number of missing values is relevant and need to address later:

```
##          variable na_count na_pct
## 1           IOTF4    54541    22%
## 2            MBMI    53311    22%
## 3      bodyweight    41128    17%
## 4      bodyheight    40036    16%
## 5      emcsocmed8    32878    13%
## 6      emcsocmed9    32855    13%
## 7      emcsocmed7    32597    13%
## 8      emcsocmed6    32544    13%
## 9      emcsocmed5    32443    13%
## 10     emcsocmed4    32428    13%
## 11     emcsocmed3    32009    13%
## 12     emcsocmed1    31973    13%
## 13     emcsocmed2    31659    13%
## 14    emconlpref3    31097    13%
## 15    emconlpref2    30586    13%
## 16    emconlpref1    29813    12%
## 17    emconlfreq2    29552    12%
## 18    emconlfreq4    29003    12%
## 19    emconlfreq3    28973    12%
## 20    emconlfreq1    26720    11%
## 21         famdec    20237     8%
## 22         famsup    20091     8%
## 23        famtalk    20019     8%
## 24       alc30d_2    19326     8%
## 25        famhelp    18896     8%
## 26    teachercare    18340     8%
## 27   teachertrust    17839     7%
## 28     talkfather    17491     7%
## 29     studaccept    17444     7%
## 30    beenbullied    16962     7%
## 31  teacheraccept    16764     7%
## 32    studhelpful    16746     7%
## 33   studtogether    16351     7%
```

```
## 34      talkmother    15521      6%
## 35    cbeenbullied    15084      6%
## 36           IRFAS    12599      5%
## 37    IRRELFAS_LMH    12599      5%
## 38 sleepdificulty    10814      4%
## 39     friendshare    10813      4%
## 40      friendtalk    10808      4%
## 41   friendcounton    10374      4%
## 42      friendhelp     9744      4%
## 43           dizzy     8566      4%
## 44         feellow     6696      3%
## 45        backache     6543      3%
## 46       irritable     6539      3%
## 47         nervous     6525      3%
## 48       physact60     6215      3%
## 49     stomachache     5679      2%
## 50 schoolpressure      5486      2%
## 51      likeschool     5109      2%
## 52         timeexe     4799      2%
## 53         lifesat     4512      2%
## 54        headache     4505      2%
## 55          health     2577      1%
## 56          agecat     1516      1%
## 57       seqno_int        0      0%
## 58       countryno        0      0%
## 59             sex        0      0%
```

Now I explore all variables by group and feature engineer as appropriate.

[1] "Individual Information"

11 vars here. So do 1x1.

```
##  [1] "seqno_int"    "countryno"    "agecat"       "sex"          "MBMI"
##  [6] "IRFAS"        "IRRELFAS_LMH" "IOTF4"        "bodyweight"   "bodyheight"
```

seqno_int is just an identifier.

countryno is recoded to country_name and this is the distribution.

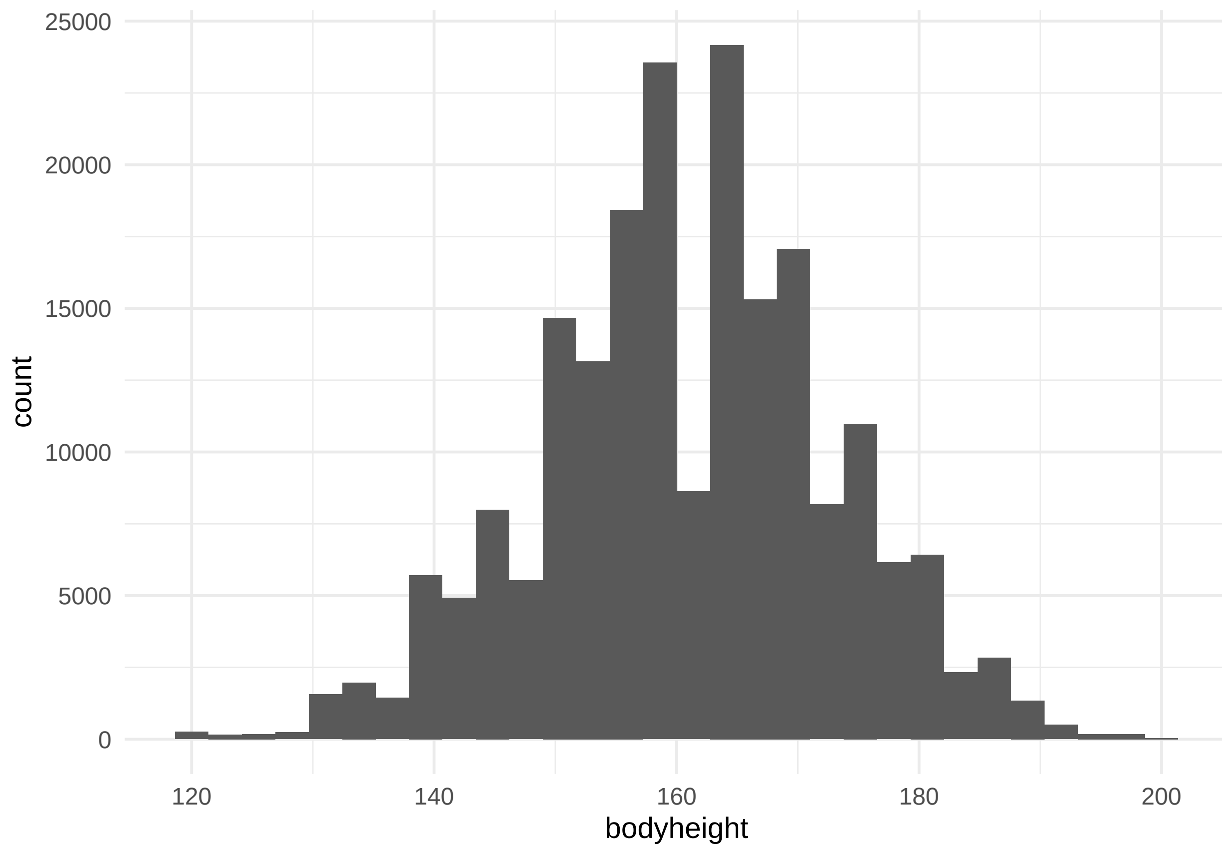
Table 2: #/% of observations by country

country_name	n	pct
Albania	1765	1%
Armenia	4717	2%
Austria	4129	2%
Azerbaijan	4586	2%
Belgium (Flemish)	4333	2%
Belgium (French)	5578	2%
Canada	12950	5%
Croatia	5169	2%

country_name	n	pct
Czech Republic	11564	5%
Denmark	3181	1%
England	3397	1%
Estonia	4725	2%
Finland	3146	1%
France	9170	4%
Georgia	4242	2%
Germany	4347	2%
Greenland	1243	1%
Hungary	3789	2%
Iceland	6996	3%
Ireland	3833	2%
Israel	7712	3%
Italy	4144	2%
Kazakhstan	4868	2%
Latvia	4412	2%
Lithuania	3797	2%
Luxembourg	4070	2%
Macedonia	4658	2%
Malta	2576	1%
Netherlands	4698	2%
Norway	3127	1%
Poland	5224	2%
Portugal	6126	3%
Republic of Moldova	4686	2%
Romania	4567	2%
Russia	4281	2%
Scotland	5021	2%
Serbia	3933	2%
Slovakia	4785	2%
Slovenia	5667	2%
Spain	4320	2%
Sweden	4185	2%
Switzerland	7510	3%
Turkey	5848	2%
Ukraine	6660	3%
Wales	15951	7%
NA	8411	3%

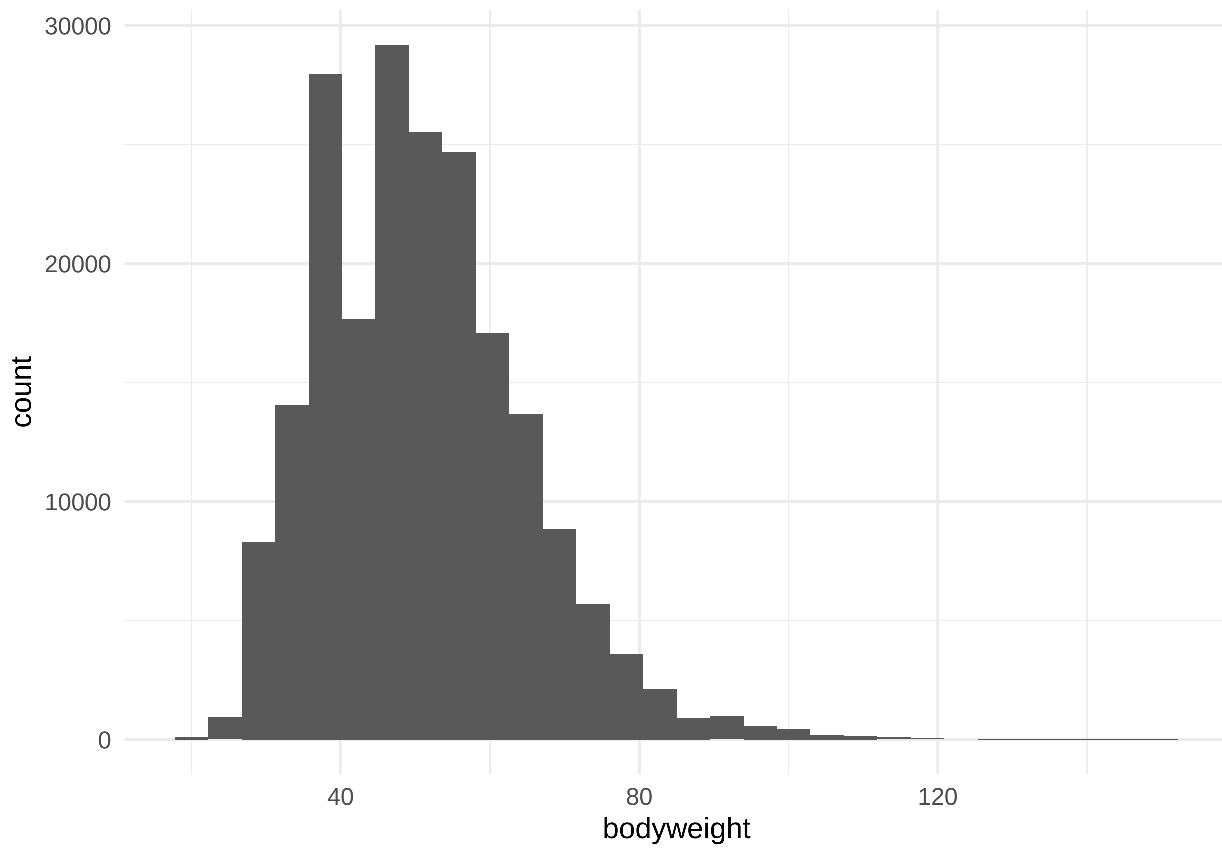
bodyheight is in cm and range from 120 to 200 so will use. Impute NAs?

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's 
##   120.0   153.0   161.0   161.1   170.0   200.0   40036
```



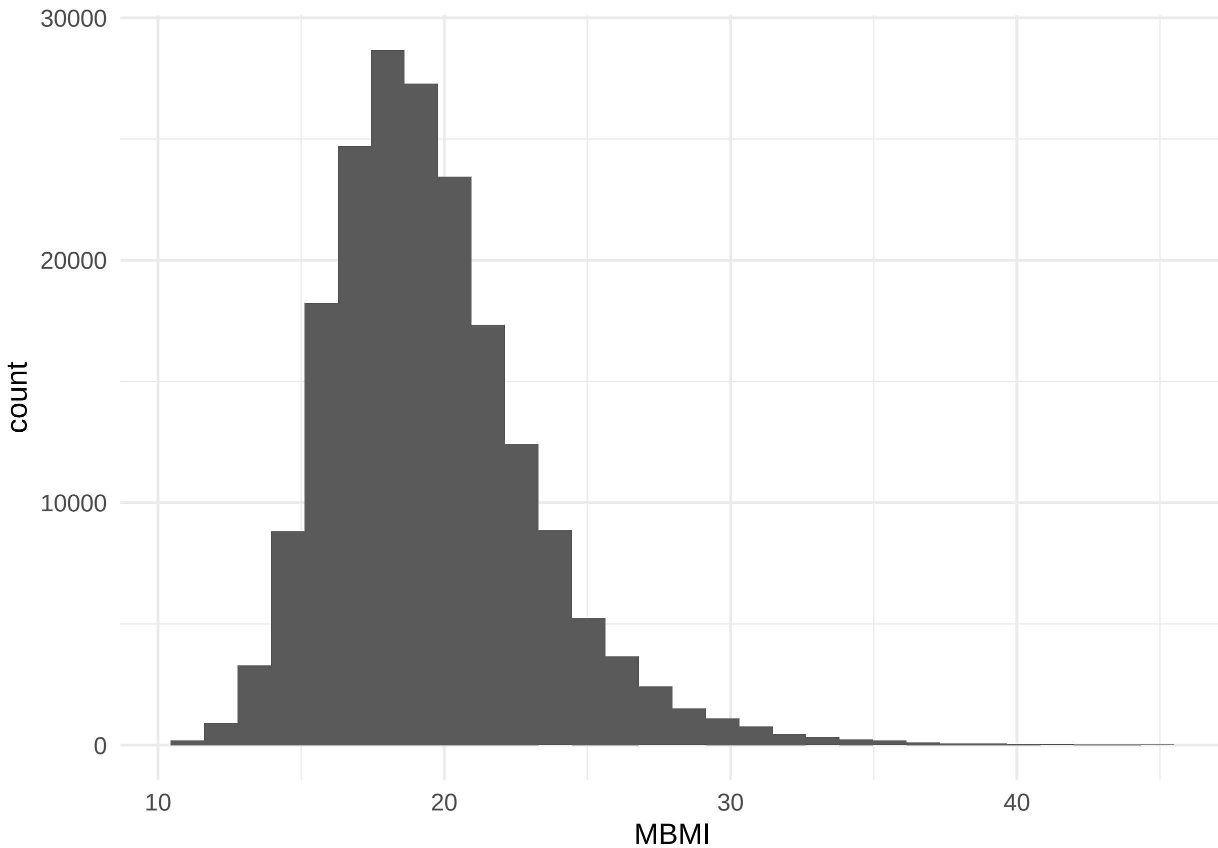
bodyweight is in kilo and range from 20 to 150 so will use maybe. Impute NAs?

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##   20.00   40.00   50.00   51.07   60.00  150.00   41128
```



MBMI is body mass index from [Range= 11.02-44.9] and 0 meaning outside overall range so make those NA.

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##    0.00   17.09   19.04   19.51   21.40   44.91   53311
```



IOTF4 is a BMI group from 1(thin) to 4(obesse) so leave as is.

```
##
##       1      2      3      4   <NA>
##   26967 132661  24487   5441  54541
```

agecat has 3 values and will leave as is: 1 ~ 11, 2 ~ 13, 3 ~ 15.

```
##
##     1     2     3  <NA>
## 81834 83749 76998  1516
```

sex is binary with 1 (boy), 2 (girl) so will leave as is.

```
##
##      1      2   <NA>
## 120350 123747      0
```

IRRELFAS_LMH is relative family affluence with 1 Low20, 2 Mid60, 3 High20. Leave as is.

IRFAS is discrete [0,13] and shows affluence.

```
##
##              1     2     3  <NA>
##    0       715     0     0     0
```

```
##   1      2238     2      0      0
##   2      4780   270      0      0
##   3      5217  3609      0      0
##   4      6829  6272      0      0
##   5      8199  9442      0      0
##   6      8522 13305    331      0
##   7      7593 17420   1513      0
##   8      1724 26873   2120      0
##   9         1 29479   2672      0
##   10        0 23830   5408      0
##   11        0 11158  11359      0
##   12        0   488  13893      0
##   13        0     0   6236      0
##   <NA>      0     0      0  12599
```

[2] "Health Issues and Life Satisfaction"

10 vars here:

```
##  [1] "health"          "lifesat"         "headache"        "stomachache"
##  [5] "backache"        "feellow"         "irritable"       "nervous"
##  [9] "sleepdificulty" "dizzy"
```

health is self rating of health from 1(Excellent) to 4(poor). Won't use for now

```
table(dat$health, useNA = "always")
```

```
##
##      1      2      3      4   <NA>
##  86985 120155  30473   3907   2577
```

lifesat asks for life satisfaction as ladder where 0(botton) to 10(best). I'll dichotomize and keep num for shap.

```
## lifesat distribution:
```

```
##
##     0     1     2     3     4     5     6     7     8     9    10  <NA>
##  1218  1041  1748  3691  6961 17137 19921 39120 55325 45271 48152  4512
```

```
## lifesat %:
```

```
##
##     0     1     2     3     4     5     6     7     8     9    10  <NA>
## 0.005 0.004 0.007 0.015 0.029 0.070 0.082 0.160 0.227 0.185 0.197 0.018
```

```
## lifesat_low as y/n:
```

```
##
##           0     1  <NA>
##   0       0  1218     0
##   1       0  1041     0
##   2       0  1748     0
##   3       0  3691     0
##   4       0  6961     0
##   5       0 17137     0
##   6   19921     0     0
##   7   39120     0     0
##   8   55325     0     0
##   9   45271     0     0
##   10  48152     0     0
##   <NA>    0     0  4512
```

headache, stomachache, backache, and dizzy are physical and will ignore for now. But many papers report on psychosomatic issues so I might come back.

feellow, irritable, nervous, sleepdificulty and mental health issues. They answer a question In the last 6 months: how often have you had the following….? And the answers goes from 1 (About every day) to 5 (Rarely or never). Their distribution is:

```
## $feellow
##
##      1      2      3      4      5
##  20224  25739  29784  48898 112756
##
## $irritable
##
##     1     2     3     4     5
## 26053 35879 42648 57922 75056
##
## $nervous
##
##     1     2     3     4     5
## 28306 32770 40064 54109 82323
##
## $sleepdificulty
##
##      1      2      3      4      5
##  30501  25543  26053  36570 114616
```

Their correlation is modest.

```
##                feellow irritable nervous sleepdificulty
## feellow           1.00      0.54    0.48           0.36
## irritable         0.54      1.00    0.53           0.35
## nervous           0.48      0.53    1.00           0.36
## sleepdificulty    0.36      0.35    0.36           1.00
```

These variables are then recoded to a binary where it flags if the complaint occurs more than once per week (values 1/2).

These frequency 1/0 vars are summed up and if sum is 2+ then there's an issue. And counts look like this:

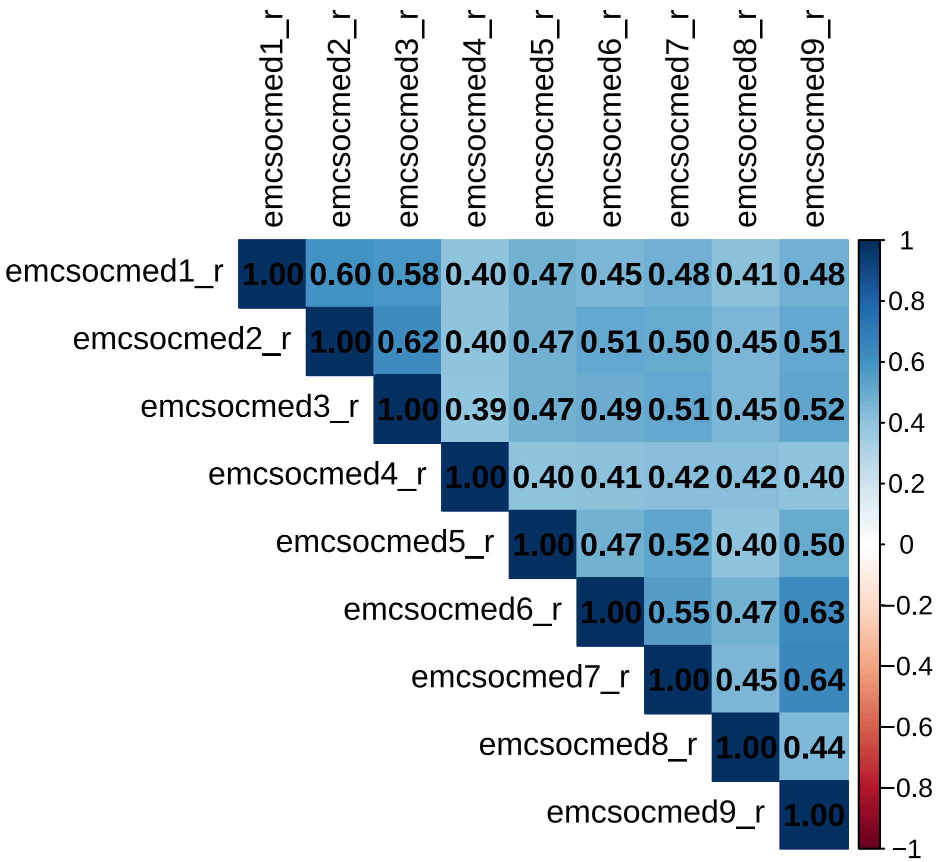
```
##
##              0      1   <NA>
##   0     121573      0      0
##   1      45509      0      0
##   2          0  29991      0
##   3          0  21172      0
##   4          0  12970      0
##   <NA>       0      0  12882
```

[3] "Problematic Social Media Use (PMSU)"

9 binary variables are recoded to 0/1.

item	topic
emcsocmed1	Social media: I can't think of anything else
emcsocmed2	Social media: Dissatisfied cause want to spend more time
emcsocmed3	Social media: Felt bad cause want to play
emcsocmed4	Social media: Failed to spend less time
emcsocmed5	Social media: Neglected other activities
emcsocmed6	Social media: Had arguments because of use
emcsocmed7	Social media: Lied about amount of time spent
emcsocmed8	Social media: Use it to escape from negative feelings
emcsocmed9	Social media: Conflict with family because of use

And they are correlated.



These 9 vars are sumed and then binned to H/M/L where H(6-9), M(2-5) and L(0-1). Resulting in:

	1	2	3	NA
0	72474	0	0	0
1	36277	0	0	0
2	0	26066	0	0
3	0	19012	0	0
4	0	14908	0	0
5	0	10306	0	0
6	0	0	5905	0
7	0	0	3382	0
8	0	0	1878	0
9	0	0	2979	0
NA	0	0	0	50910

[4] "Bullying"

There are 2 variables that measure if kid has been bullied. a) beenbullied: Been bullied past months and b) cbeenbullied: Been cyber bullied. Answers range from 1 (Haven't) to 5 (Several times a week).

I will derive 2 to keep: cbeenbullied as y/n and bulliedsum as sum of both.

```
## cbullied recoding:
```

```
##
##             0      1   <NA>
##   1    198900      0      0
##   2         0  20426      0
##   3         0   4625      0
##   4         0   2274      0
##   5         0   2788      0
##   <NA>      0      0  15084
```

```
## bullying score: (low is better)
```

score	n	pct
2	151013	62%
3	34034	14%
4	13753	6%
5	7360	3%
6	7703	3%
7	2745	1%
8	1288	1%
9	747	0%
10	1093	0%
NA	24361	10%

[5] "Communicates Online More Than Real Life"

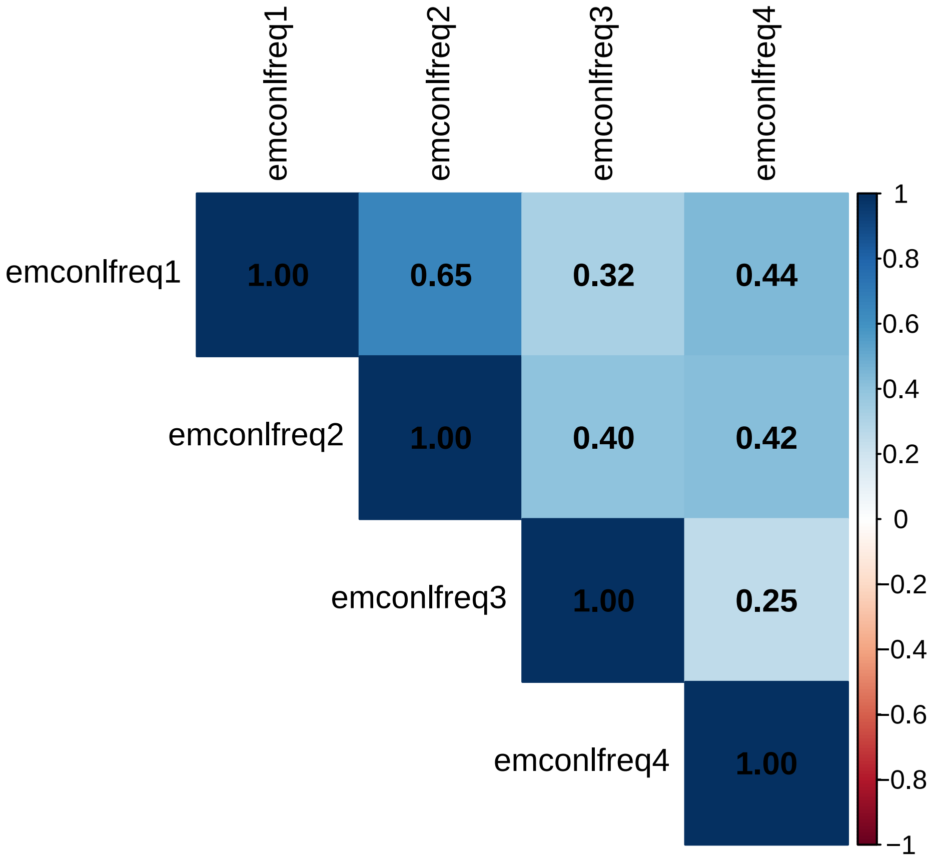
7 vars here in 2 themes - a) continuos online communications and b) preference to communicate online.

For a) continuos online communications, measures how how often young person communicate online with 4 types of people in their life. Answers range from 1 (NA or don't know) to 6 (almost all of the time).

Variable	Description
emconlfreq1	Online contact with close friends
emconlfreq2	Online contact with larger friend group
emconlfreq3	Online contact with online friends
emconlfreq4	Online contact with other

And they are somewhat correlated but not much.

```
## [1] "emconlfreq1" "emconlfreq2" "emconlfreq3" "emconlfreq4" "emconlpref1"
## [6] "emconlpref2" "emconlpref3"
```

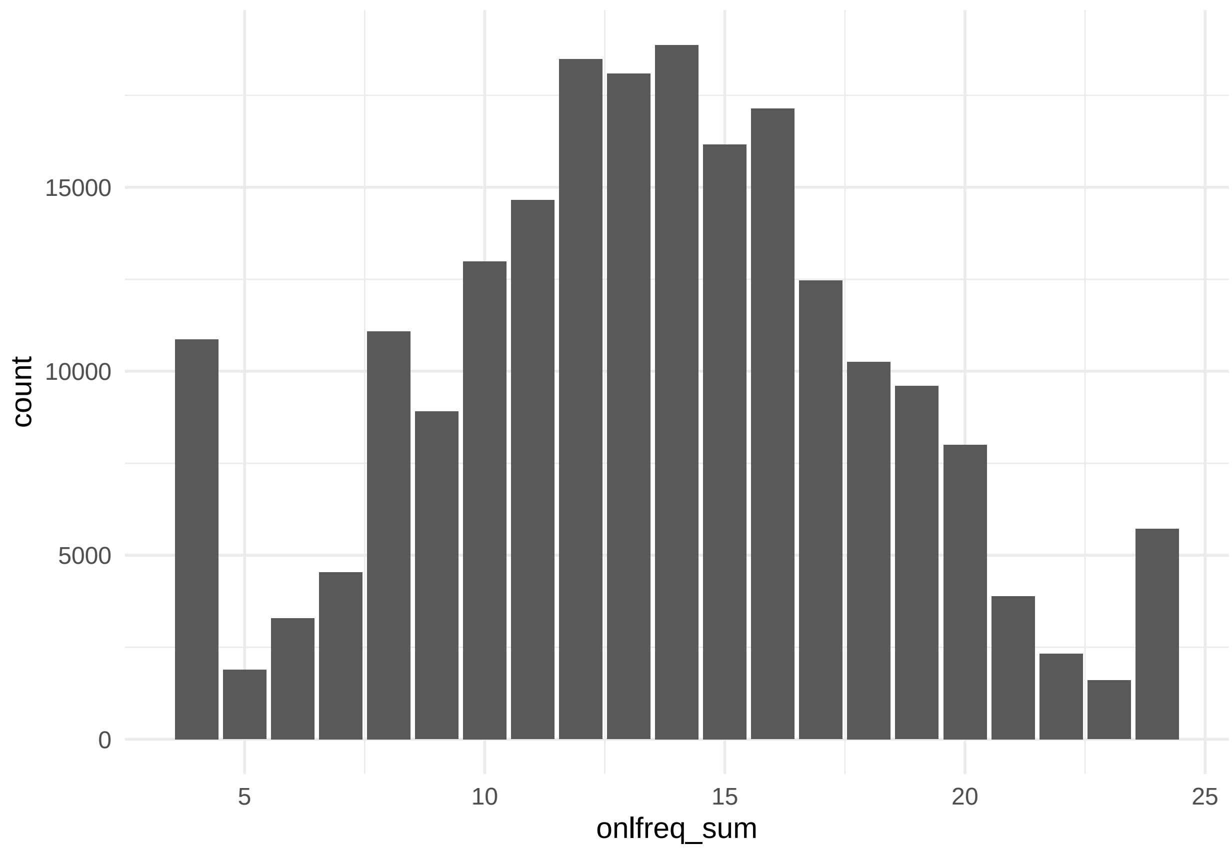


I combine them into two options. a) one dichotomized and adopt the HBSC definition of yes if any of the variables is 6; and b) a sum for our shap tests with values in [4,24].

```
## Online frequence is (onlfreq_is):
```

```
##
##      0      1   <NA>
## 143585  74898  25614
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##    4.00   10.00   14.00   13.51   17.00   24.00   33251
```

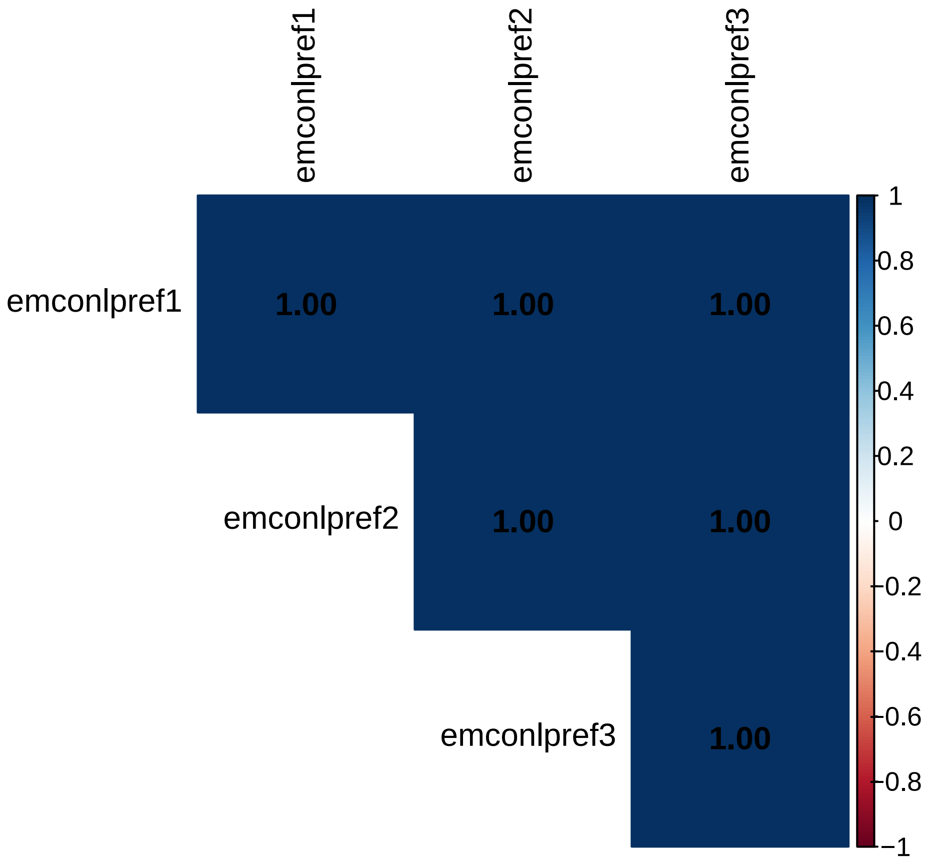


For b) preference to communicate online, 3 questions ask about preference of sharing important stuff online vs real-life. The three variables are:

Variable	Description
emconlpref1	Secrets, more easily online
emconlpref2	Feelings, more easily online
emconlpref3	Concerns, more easily online

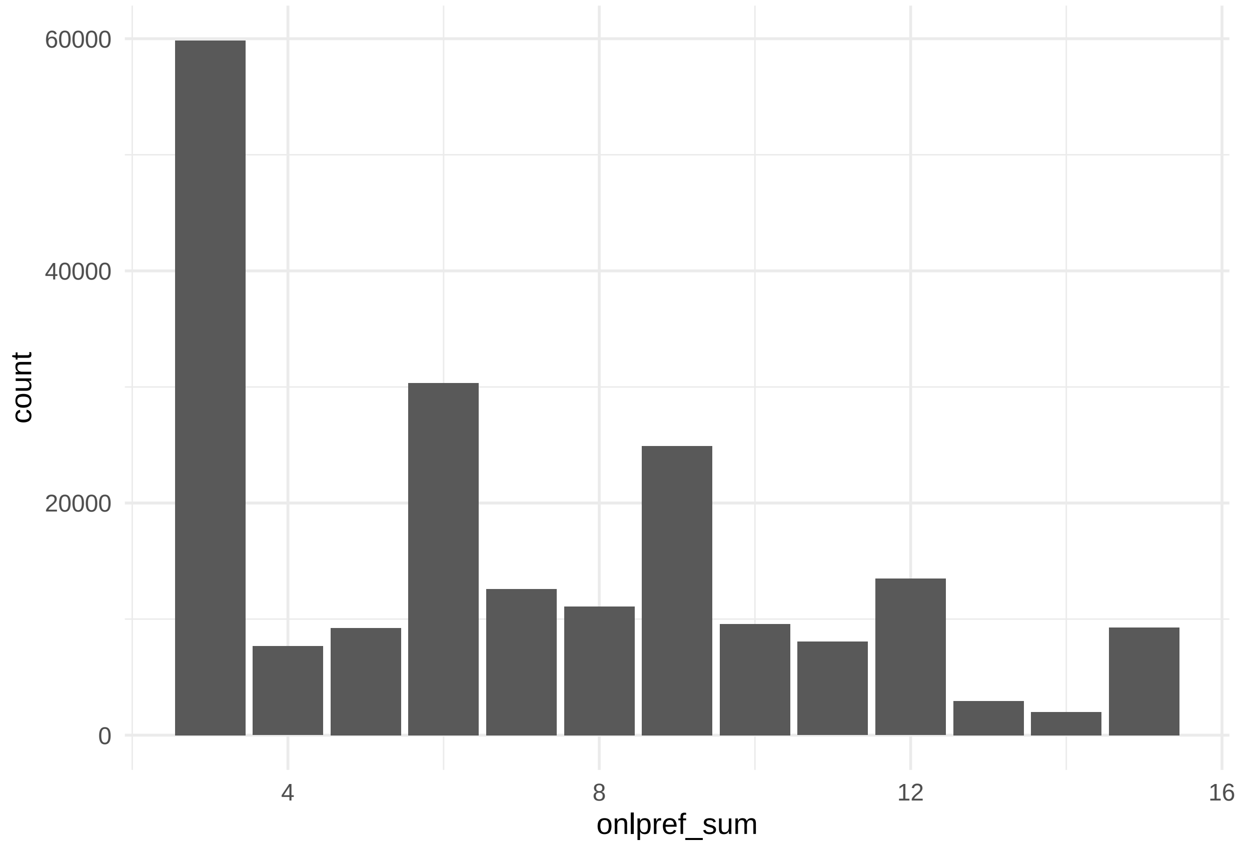
The answers range from 1 (Strong disagree) to 5 (Strong agree).

And they are correlated. Remove 99s????



Haven't found a paper that uses them so I will replicated method of friends support below. Summing them results in [3,15] and then grouped as H/M/L as L 3-6, M 7-10, H 11-15. Resulting in:

```
## onlpref:

##
##      1      2      3   <NA>
## 107085  58134  35713  43165

##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##   3.000   3.000   6.000   6.919   9.000  15.000   43165
```



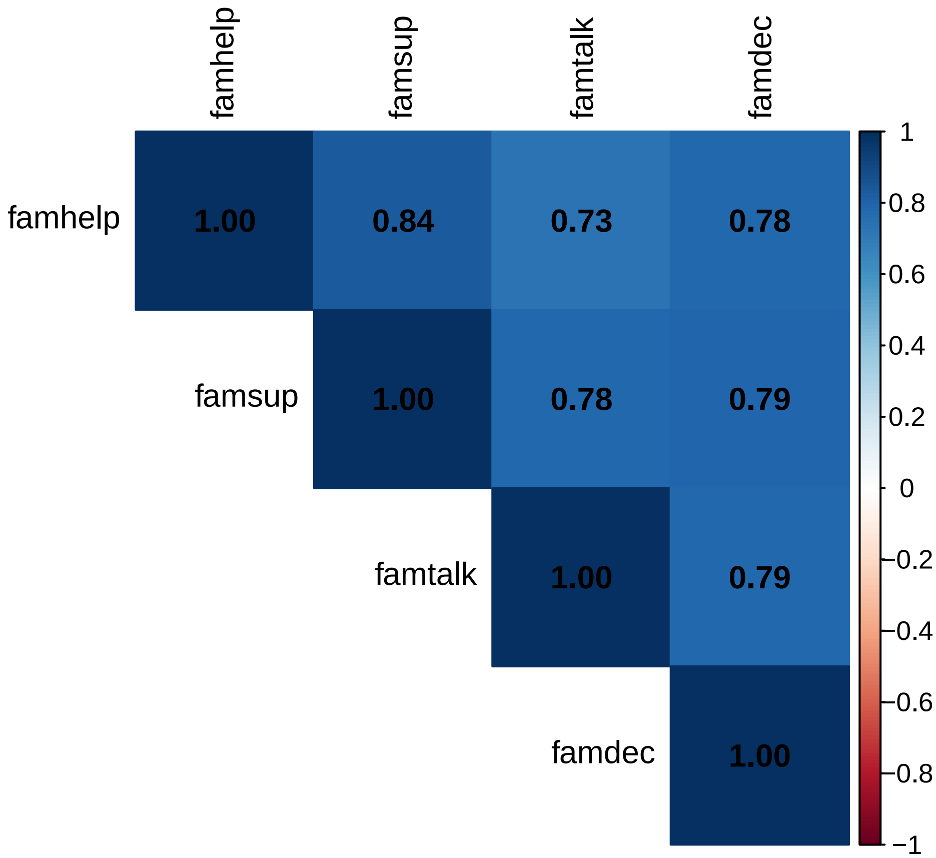
[6] "Family Support"

4 questions going from 1 Very strongly disagree to 7 Very strongly agree.

```
## [1] "famhelp" "famsup"  "famtalk" "famdec"
```

item	topic
famhelp	Family tries to help
famsup	Get emotional help from family
famtalk	Talk about problems with family
famclose	Feel close to family

And they are correlated.



Summing them results in [4,28] and then grouped as H/M/L as L 4-11, M 12-19, H 20-28. Resulting in:

	1	2	3	NA
4	10939	0	0	0
5	1597	0	0	0
6	1452	0	0	0
7	1382	0	0	0
8	2477	0	0	0
9	1485	0	0	0
10	2274	0	0	0
11	1739	0	0	0
12	0	2655	0	0
13	0	2279	0	0
14	0	2745	0	0
15	0	2781	0	0
16	0	5747	0	0
17	0	3440	0	0
18	0	4455	0	0
19	0	4637	0	0
20	0	0	7739	0
21	0	0	5853	0
22	0	0	9240	0
23	0	0	7878	0
24	0	0	13474	0
25	0	0	11089	0

	1	2	3	NA
26	0	0	13090	0
27	0	0	15938	0
28	0	0	85359	0
NA	0	0	0	22353

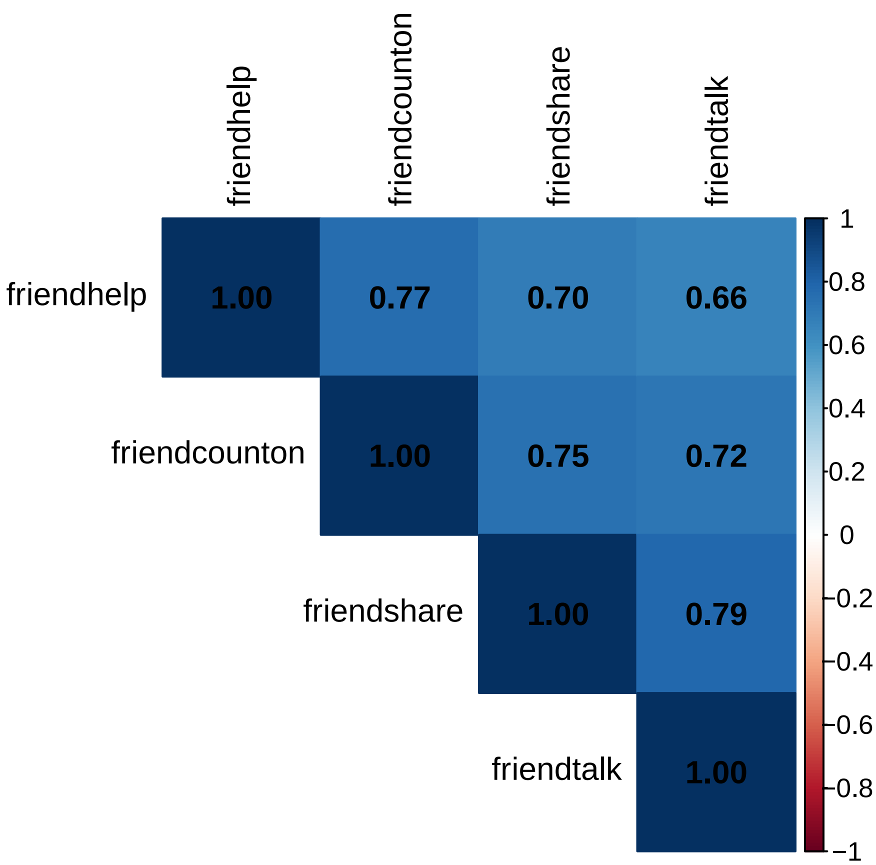
[7] "Friends Support"

4 questions going from 1 Very strongly disagree to 7 Very strongly agree.

```
## [1] "friendhelp"    "friendcounton" "friendshare"   "friendtalk"
```

item	topic
friendhelp	My friends really try to help me
friendcounton	I can count on my friends when things go wrong
friendshare	I have friends with whom I can share my joys and sorrows
friendtalk	I can talk about my problems with my friends

And they are correlated.



Summing them results in [4,28] and then grouped as H/M/L as L 4-11, M 12-19, H 20-28. Resulting in:

	1	2	3	NA
4	9995	0	0	0
5	2159	0	0	0
6	2109	0	0	0
7	2392	0	0	0
8	3798	0	0	0
9	2463	0	0	0
10	3752	0	0	0
11	2869	0	0	0
12	0	4267	0	0
13	0	3834	0	0
14	0	4428	0	0
15	0	4339	0	0
16	0	9075	0	0
17	0	5750	0	0
18	0	7126	0	0
19	0	7644	0	0
20	0	0	11463	0
21	0	0	8358	0
22	0	0	12583	0
23	0	0	10381	0
24	0	0	16855	0
25	0	0	12468	0
26	0	0	12977	0
27	0	0	14317	0
28	0	0	55799	0
NA	0	0	0	12896

[8] "School Support"

8 variables related to school.

```
## $`School Support`
## [1] "likeschool"     "schoolpressure" "studtogether"   "studhelpful"
## [5] "studaccept"     "teacheraccept"  "teachercare"    "teachertrust"
```

likeschool ask if they like school and goes from 1 (lots) to 4 (no). I won't use for now.

schoolpressure ask if they feel pressure and goes from 1(no) to 4(a lot). I won't use for now.

Then there are 3 questions about relationship with students and 3 with relationship with teachers. The answers go from 1 (strong agree) to 5 (strong disagree).

category	item	topic
teacher	teacheraccept	Teacher accepts me
teacher	teachercare	Teacher cares about me
teacher	teachertrust	Feel trust in teacher
classmates	studtogether	Students enjoy being together
classmates	studhelpful	Students kind and helpful
classmates	studaccept	Students accept me

Scales are reversed so 1 is the negative and 5 is the positive.

And there is moderate correlation between groups but lowers across stud/teach.

```
##                 teacheraccept_r teachercare_r teachertrust_r studtogether_r
## teacheraccept_r            1.00          0.60           0.59           0.28
## teachercare_r              0.60          1.00           0.65           0.27
## teachertrust_r             0.59          0.65           1.00           0.28
## studtogether_r             0.28          0.27           0.28           1.00
## studhelpful_r              0.30          0.30           0.32           0.56
## studaccept_r               0.35          0.27           0.27           0.45
##                 studhelpful_r studaccept_r
## teacheraccept_r          0.30         0.35
## teachercare_r            0.30         0.27
## teachertrust_r           0.32         0.27
## studtogether_r           0.56         0.45
## studhelpful_r            1.00         0.54
## studaccept_r             0.54         1.00
```

A few features are derived. For stud/teach separate a binary following HBSC and a sum for shap test.

```
## teachersup:
```

```
##
##            0     1  <NA>
##   3     3593     0     0
##   4     1483     0     0
##   5     2657     0     0
##   6     5531     0     0
##   7     6727     0     0
##   8     9789     0     0
##   9    19836     0     0
##   10   22972     0     0
##   11   27699     0     0
##   12       0 40136     0
##   13       0 23751     0
##   14       0 21349     0
##   15       0 38813     0
##   <NA>     0     0 19761
```

```
## studentsup:
```

```
##
##            0     1  <NA>
##   3     2138     0     0
##   4     1199     0     0
##   5     2018     0     0
##   6     4392     0     0
##   7     6367     0     0
##   8     9336     0     0
##   9    18744     0     0
##   10   22280     0     0
##   11   30233     0     0
```

```
##   12        0 48125     0
##   13        0 26851     0
##   14        0 21371     0
##   15        0 32245     0
##   <NA>      0     0 18798
```

[9] "Communication with Parents"

Two vars with answers from 1 (Very easy) to 5 (Don't have or see). With this counts and proportions in data.

```
## [1] "talkfather" "talkmother"
```

```
##           talkmother
## talkfather     1     2     3     4     5  <NA>
##       1    62315  6269  2345   842   891  1115
##       2    31505 40099  4150   959   753   880
##       3     9936 17764 10530  1130   499   430
##       4     3719  5070  4602  5132   294   193
##       5     5842  4188  1968  1039  1905   242
##       <NA>  2582  1523   447   174   104 12661
```

```
##           talkmother
## talkfather     1     2     3     4     5  <NA>
##       1    0.255 0.026 0.010 0.003 0.004 0.005
##       2    0.129 0.164 0.017 0.004 0.003 0.004
##       3    0.041 0.073 0.043 0.005 0.002 0.002
##       4    0.015 0.021 0.019 0.021 0.001 0.001
##       5    0.024 0.017 0.008 0.004 0.008 0.001
##       <NA> 0.011 0.006 0.002 0.001 0.000 0.052
```

With some correlation between them (I expected more).

So I will dichotomize like HBSC and keep sum of both as well for SHAP. These are dichotomized for each parent as Yes if [1,2] and No if [3-5].

Where new vars look like this for dichotom values:

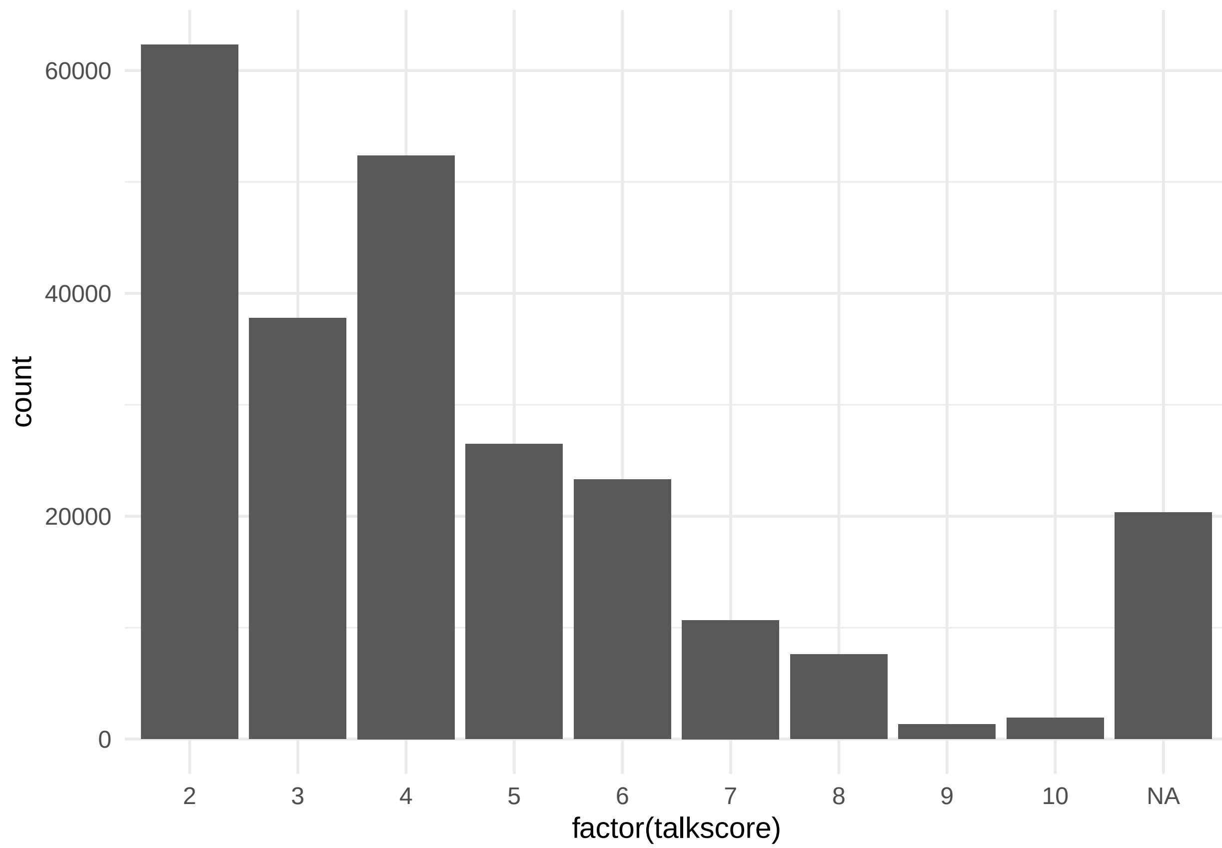
```
## Father:
```

```
##
##            0     1  <NA>
##   1        0 73777     0
##   2        0 78346     0
##   3    40289     0     0
##   4    19010     0     0
##   5    15184     0     0
##   <NA>     0     0 17491
```

```
## Mother:
```

```
##
##          0     1  <NA>
##   1      0 73777     0
##   2      0 78346     0
##   3  40289     0     0
##   4  19010     0     0
##   5  15184     0     0
##   <NA>   0     0 17491
```

And the talk score looks like this. Will keep for now and orig vars as well.



[10] "Good/Bad Habits"

physact60 is just past week so ignore.

timeexe ask how often exercise in free time going from 1 (every day) to 7 never. HBSC shows proportions of those that do 3+ per week. But that is 2022 data, in our case we do 2-3 times per week or more. And I keep timexe as is for shap test.

```
##
##          0     1  <NA>
##   1      0 41797     0
##   2      0 56673     0
##   3      0 71661     0
##   4  32246     0     0
```

```
## 5    10541     0     0
## 6    10256     0     0
## 7    16124     0     0
## <NA>     0     0  4799
```

alc30d_2 ask about alcohol use in past 30 days and will ignore for now

Clean Dataset for Modeling

Missing Values

Need to plan for this. For now, just keeping complete obs, but note some countries are very affected.

country_name	n	na_FALSE	pct_complete
Portugal	6126	4955	0.809
Finland	3146	2351	0.747
Estonia	4725	3470	0.734
Hungary	3789	2779	0.733
Italy	4144	3007	0.726
Lithuania	3797	2721	0.717
Austria	4129	2923	0.708
Slovenia	5667	3974	0.701
Republic of Moldova	4686	3230	0.689
Spain	4320	2964	0.686
Czech Republic	11564	7877	0.681
Serbia	3933	2602	0.662
Belgium (Flemish)	4333	2836	0.655
Ukraine	6660	4341	0.652
Netherlands	4698	3058	0.651
Germany	4347	2824	0.650
Iceland	6996	4430	0.633
Russia	4281	2604	0.608
Albania	1765	1056	0.598
Turkey	5848	3466	0.593
Croatia	5169	2934	0.568
Sweden	4185	2353	0.562
France	9170	5056	0.551
Romania	4567	2500	0.547
Luxembourg	4070	2171	0.533
Israel	7712	4027	0.522
Georgia	4242	2127	0.501
Norway	3127	1472	0.471
Kazakhstan	4868	2179	0.448
Canada	12950	5591	0.432
Armenia	4717	1950	0.413
Belgium (French)	5578	2207	0.396
Malta	2576	612	0.238
Scotland	5021	1068	0.213
Wales	15951	2832	0.178
Ireland	3833	603	0.157
England	3397	254	0.075

country_name	n	na_FALSE	pct_complete
Azerbaijan	4586	0	0.000
Denmark	3181	0	0.000
Greenland	1243	0	0.000
Latvia	4412	0	0.000
Macedonia	4658	0	0.000
Poland	5224	0	0.000
Slovakia	4785	0	0.000
Switzerland	7510	0	0.000
NA	8411	0	0.000

Data Type Coercions

```r
# # vars to coerce to fact (i looked 1x1)
# country_name
# IOTF4 ordered 1-4
# agecat ordered 1-3
# sex
# IRRELFAS_LMH ordered 1-3
# lifesat_low ordered 0-1
# mental_issue ordered 0-1
# pmsu_lmh order 1-3
# cbullied order 0-1
# onlfreq_is order 0-1
# onlpref order 1-3
# famsup_hml order 1-3
# frisup_hml order 1-3
# teachersup order 0-1
# studsup order 0-1
# talkf order 0-1
# talkm order 0-1
# timeexe_r order 0-1

d <- d %>%
  mutate(
    # --- Unordered (Nominal) Factors ---
    country_name = as.factor(country_name),
    sex          = as.factor(sex),
    # --- Ordered Factors (1 to 4) ---
    IOTF4        = factor(IOTF4, levels = c("1", "2", "3", "4"), ordered = TRUE),
    # --- Ordered Factors (1 to 3) ---
    agecat       = factor(agecat,       levels = c("1", "2", "3"), ordered = TRUE),
    IRRELFAS_LMH = factor(IRRELFAS_LMH, levels = c("1", "2", "3"), ordered = TRUE),
    pmsu_lmh     = factor(pmsu_lmh,     levels = c("1", "2", "3"), ordered = TRUE),
    onlpref      = factor(onlpref,      levels = c("1", "2", "3"), ordered = TRUE),
    famsup_hml   = factor(famsup_hml,   levels = c("1", "2", "3"), ordered = TRUE),
    frisup_hml   = factor(frisup_hml,   levels = c("1", "2", "3"), ordered = TRUE),
    # --- Ordered Factors (0 to 1 Binary / Dummy Scales) ---
    lifesat_low  = factor(lifesat_low,  levels = c("0", "1"), ordered = TRUE),
    mental_issue = factor(mental_issue, levels = c("0", "1"), ordered = TRUE),
```

```
    cbullied     = factor(cbullied,     levels = c("0", "1"), ordered = TRUE),
    onlfreq_is   = factor(onlfreq_is,   levels = c("0", "1"), ordered = TRUE),
    teachersup   = factor(teachersup,   levels = c("0", "1"), ordered = TRUE),
    studsup      = factor(studsup,      levels = c("0", "1"), ordered = TRUE),
    talkf        = factor(talkf,        levels = c("0", "1"), ordered = TRUE),
    talkm        = factor(talkm,        levels = c("0", "1"), ordered = TRUE),
    timeexe_r    = factor(timeexe_r,    levels = c("0", "1"), ordered = TRUE)
  )
```

Tests for Potential of Prediction

```
#vars_for cochran (ordered)
#vars_for_chi
#vars_for_tt

# --- 1. SETUP YOUR TARGET VARIABLE NAME ---
# Change "depression" to your exact binary response column name
# IMPORTANT: This target variable must have exactly two levels (e.g., "No" and "Yes")
#target_var <- "depression"
target_var <- "mental_issue"
# Ensure target is a factor and identify the "success" level (the 2nd level)
#d[[target_var]] <- as.factor(d[[target_var]])
target_levels <- levels(d[[target_var]])
success_level <- target_levels[2]

# --- 2. THE AUTOMATED LOOP ---
for (col_name in names(d)) {

  # Skip the target variable itself
  if (col_name %in% c(target_var, "seqno_int")) next

  cat("\n==================================================\n")
  cat("Testing Relationship:", target_var, "vs", col_name, "\n")
  cat("==================================================\n")

  # --- 1. IF ORDINAL -> RUN COCHRAN-ARMITAGE TREND TEST ---
  if (is.ordered(d[[col_name]])) {
    cat("Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...\n\n")

    # Build table: rows = target, columns = ordinal levels
    contingency_table <- table(d[[target_var]], d[[col_name]])
    print(contingency_table)
    cat("\n")

    # Extract successes for the 2nd level (e.g., "Yes") and column totals
    successes <- contingency_table[success_level, ]
    totals    <- colSums(contingency_table)

    test_result <- tryCatch({
      prop.trend.test(successes, totals)
    }, error = function(e) { return(NULL) })
```

```r
    if (!is.null(test_result)) {
      print(test_result)
    } else {
      cat("Error: Could not run Trend Test. Verify your variable setup.\n")
    }

  # --- 2. IF NUMERIC -> RUN T-TEST ---
  } else if (is.numeric(d[[col_name]])) {
    cat("Predictor Type: NUMERIC -> Running Independent t-test...\n\n")

    test_result <- tryCatch({
      t.test(d[[col_name]] ~ d[[target_var]])
    }, error = function(e) { return(NULL) })

    if (!is.null(test_result)) {
      print(test_result)
    } else {
      cat("Error: Could not run t-test. Check for missing values.\n")
    }

  # --- 3. IF NOMINAL / BINARY -> RUN CHI-SQUARE ---
  } else if (is.factor(d[[col_name]]) || is.character(d[[col_name]])) {
    cat("Predictor Type: NOMINAL/BINARY -> Running Chi-Square Test...\n\n")

    contingency_table <- table(d[[target_var]], d[[col_name]])
    print(contingency_table)
    cat("\n")

    test_result <- tryCatch({
      chisq.test(contingency_table)
    }, error = function(e) { return(NULL) })

    if (!is.null(test_result)) {
      print(test_result)
    } else {
      cat("Error: Could not run Chi-Square test.\n")
    }
  }
}
```

```
##
## ==================================================
## Testing Relationship: mental_issue vs country_name
## ==================================================
## Predictor Type: NOMINAL/BINARY -> Running Chi-Square Test...
##
##
##     Albania Armenia Austria Belgium (Flemish) Belgium (French) Canada Croatia
##   0     790    1361    2296              2247             1513   4124    2266
##   1     266     589     627               589              694   1467     668
##
##     Czech Republic England Estonia Finland France Georgia Germany Hungary
##   0           5744     159    2409    1769   3526    1660    2328    2010
```

```
##   1           2133      95    1061     582   1530     467     496     769
##
##     Iceland Ireland Israel Italy Kazakhstan Lithuania Luxembourg Malta
##   0    3309     432   2582  1739       1850      2059       1564   404
##   1    1121     171   1445  1268        329       662        607   208
##
##     Netherlands Norway Portugal Republic of Moldova Romania Russia Scotland
##   0        2491   1211     3750                2388    1681   2012      787
##   1         567    261     1205                 842     819    592      281
##
##     Serbia Slovenia Spain Sweden Turkey Ukraine Wales
##   0   1984     3087  2415   1581   1640    2915  2015
##   1    618      887   549    772   1826    1426   817
##
##
##  Pearson's Chi-squared test
##
## data:  contingency_table
## X-squared = 2776.1, df = 36, p-value < 2.2e-16
##
##
## ==================================================
## Testing Relationship: mental_issue vs bodyheight
## ==================================================
## Predictor Type: NUMERIC -> Running Independent t-test...
##
##
##  Welch Two Sample t-test
##
## data:  d[[col_name]] by d[[target_var]]
## t = -7.5384, df = 57278, p-value = 4.828e-14
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##  -0.7394680 -0.4342877
## sample estimates:
## mean in group 0 mean in group 1
##        161.7640        162.3509
##
##
## ==================================================
## Testing Relationship: mental_issue vs bodyweight
## ==================================================
## Predictor Type: NUMERIC -> Running Independent t-test...
##
##
##  Welch Two Sample t-test
##
## data:  d[[col_name]] by d[[target_var]]
## t = -20.023, df = 53178, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##  -2.017091 -1.657400
## sample estimates:
## mean in group 0 mean in group 1
```

```
##        51.55489        53.39214
##
##
## ==================================================
## Testing Relationship: mental_issue vs MBMI
## ==================================================
## Predictor Type: NUMERIC -> Running Independent t-test...
##
##
##  Welch Two Sample t-test
##
## data:  d[[col_name]] by d[[target_var]]
## t = -24.228, df = 49184, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##  -0.6638405 -0.5644728
## sample estimates:
## mean in group 0 mean in group 1
##        19.45514        20.06930
##
##
## ==================================================
## Testing Relationship: mental_issue vs IOTF4
## ==================================================
## Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...
##
##
##         1     2     3     4
##   0 10993 55631  9539  1935
##   1  4051 20106  4076  1073
##
##
##  Chi-squared Test for Trend in Proportions
##
## data:  successes out of totals ,
##  using scores: 1 2 3 4
## X-squared = 105.21, df = 1, p-value < 2.2e-16
##
##
## ==================================================
## Testing Relationship: mental_issue vs agecat
## ==================================================
## Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...
##
##
##         1     2     3
##   0 23818 27346 26934
##   1  6269 10137 12900
##
##
##  Chi-squared Test for Trend in Proportions
##
## data:  successes out of totals ,
##  using scores: 1 2 3
```

```
## X-squared = 1151.5, df = 1, p-value < 2.2e-16
##
##
## ==================================================
## Testing Relationship: mental_issue vs sex
## ==================================================
## Predictor Type: NOMINAL/BINARY -> Running Chi-Square Test...
##
##
##         1     2
##   0 40500 37598
##   1 10256 19050
##
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  contingency_table
## X-squared = 2430.1, df = 1, p-value < 2.2e-16
##
##
## ==================================================
## Testing Relationship: mental_issue vs IRRELFAS_LMH
## ==================================================
## Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...
##
##
##         1     2     3
##   0 13352 49163 15583
##   1  5728 18149  5429
##
##
##  Chi-squared Test for Trend in Proportions
##
## data:  successes out of totals ,
##  using scores: 1 2 3
## X-squared = 85.88, df = 1, p-value < 2.2e-16
##
##
## ==================================================
## Testing Relationship: mental_issue vs IRFAS
## ==================================================
## Predictor Type: NUMERIC -> Running Independent t-test...
##
##
##  Welch Two Sample t-test
##
## data:  d[[col_name]] by d[[target_var]]
## t = 19.655, df = 52226, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##  0.3289788 0.4018589
## sample estimates:
## mean in group 0 mean in group 1
##        8.082576        7.717157
```

```
##
##
## ====================================================
## Testing Relationship: mental_issue vs lifesat_low
## ====================================================
## Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...
##
##
##         0     1
##   0 73232  4866
##   1 21294  8012

##
##  Chi-squared Test for Trend in Proportions
##
## data:  successes out of totals ,
##  using scores: 1 2
## X-squared = 8997.7, df = 1, p-value < 2.2e-16
##
##
## ====================================================
## Testing Relationship: mental_issue vs mental_sum
## ====================================================
## Predictor Type: NUMERIC -> Running Independent t-test...
##
##
##  Welch Two Sample t-test
##
## data:  d[[col_name]] by d[[target_var]]
## t = -518.82, df = 36694, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##  -2.490436 -2.471690
## sample estimates:
## mean in group 0 mean in group 1
##       0.2641425       2.7452058
##
##
## ====================================================
## Testing Relationship: mental_issue vs pmsu_lmh
## ====================================================
## Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...
##
##
##         1     2     3
##   0 49970 24610  3518
##   1 11592 14012  3702
##
##
##  Chi-squared Test for Trend in Proportions
##
## data:  successes out of totals ,
##  using scores: 1 2 3
## X-squared = 5875.6, df = 1, p-value < 2.2e-16
```

```
##
##
## ====================================================
## Testing Relationship: mental_issue vs pmsu_sum
## ====================================================
## Predictor Type: NUMERIC -> Running Independent t-test...
##
##
##  Welch Two Sample t-test
##
## data:  d[[col_name]] by d[[target_var]]
## t = -75.819, df = 44860, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##  -1.176923 -1.117607
## sample estimates:
## mean in group 0 mean in group 1
##        1.474109        2.621374
##
##
## ====================================================
## Testing Relationship: mental_issue vs cbullied
## ====================================================
## Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...
##
##
##         0     1
##   0 71095  7003
##   1 23089  6217

##
##  Chi-squared Test for Trend in Proportions
##
## data:  successes out of totals ,
##  using scores: 1 2
## X-squared = 2961.3, df = 1, p-value < 2.2e-16
##
##
## ====================================================
## Testing Relationship: mental_issue vs bulliedsum
## ====================================================
## Predictor Type: NUMERIC -> Running Independent t-test...
##
##
##  Welch Two Sample t-test
##
## data:  d[[col_name]] by d[[target_var]]
## t = -57.626, df = 38594, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##  -0.6125673 -0.5722677
## sample estimates:
## mean in group 0 mean in group 1
##        2.453955        3.046373
```

```
##
##
## ====================================================
## Testing Relationship: mental_issue vs onlfreq_is
## ====================================================
## Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...
##
##
##         0     1
##   0 52129 25969
##   1 16484 12822

##
##  Chi-squared Test for Trend in Proportions
##
## data:  successes out of totals ,
##  using scores: 1 2
## X-squared = 1018.3, df = 1, p-value < 2.2e-16
##
##
## ====================================================
## Testing Relationship: mental_issue vs onlfreq_sum
## ====================================================
## Predictor Type: NUMERIC -> Running Independent t-test...
##
##
##  Welch Two Sample t-test
##
## data:  d[[col_name]] by d[[target_var]]
## t = -21.652, df = 51787, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##  -0.6664252 -0.5557844
## sample estimates:
## mean in group 0 mean in group 1
##        13.95915        14.57026
##
##
## ====================================================
## Testing Relationship: mental_issue vs onlpref
## ====================================================
## Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...
##
##
##         1     2     3
##   0 44213 22098 11787
##   1 13274  8743  7289
##
##
##  Chi-squared Test for Trend in Proportions
##
## data:  successes out of totals ,
##  using scores: 1 2 3
## X-squared = 1621.5, df = 1, p-value < 2.2e-16
```

```
##
##
## ===================================================
## Testing Relationship: mental_issue vs onlpref_sum
## ===================================================
## Predictor Type: NUMERIC -> Running Independent t-test...
##
##
##  Welch Two Sample t-test
##
## data:  d[[col_name]] by d[[target_var]]
## t = -40.243, df = 48583, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##  -1.0689588 -0.9696696
## sample estimates:
## mean in group 0 mean in group 1
##        6.609260        7.628574
##
##
## ===================================================
## Testing Relationship: mental_issue vs famsup_hml
## ===================================================
## Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...
##
##
##         1     2     3
##   0  5903  6733 65462
##   1  3837  6165 19304
##
##
##  Chi-squared Test for Trend in Proportions
##
## data:  successes out of totals ,
##  using scores: 1 2 3
## X-squared = 2998.4, df = 1, p-value < 2.2e-16
##
##
## ===================================================
## Testing Relationship: mental_issue vs famsup_sum
## ===================================================
## Predictor Type: NUMERIC -> Running Independent t-test...
##
##
##  Welch Two Sample t-test
##
## data:  d[[col_name]] by d[[target_var]]
## t = 61.311, df = 47945, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##  2.803933 2.989129
## sample estimates:
## mean in group 0 mean in group 1
##        23.86316        20.96663
```

```
##
##
## ====================================================
## Testing Relationship: mental_issue vs frisup_hml
## ====================================================
## Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...
##
##
##         1     2     3
##   0  7241 13510 57347
##   1  4175  6915 18216
##
##
##  Chi-squared Test for Trend in Proportions
##
## data:  successes out of totals ,
##  using scores: 1 2 3
## X-squared = 1241.3, df = 1, p-value < 2.2e-16
##
##
## ====================================================
## Testing Relationship: mental_issue vs frisup_sum
## ====================================================
## Predictor Type: NUMERIC -> Running Independent t-test...
##
##
##  Welch Two Sample t-test
##
## data:  d[[col_name]] by d[[target_var]]
## t = 35.397, df = 49170, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##  1.581894 1.767350
## sample estimates:
## mean in group 0 mean in group 1
##        22.04383        20.36921
##
##
## ====================================================
## Testing Relationship: mental_issue vs teachersup
## ====================================================
## Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...
##
##
##         0     1
##   0 32592 45506
##   1 17828 11478

##
##  Chi-squared Test for Trend in Proportions
##
## data:  successes out of totals ,
##  using scores: 1 2
## X-squared = 3121.8, df = 1, p-value < 2.2e-16
```

```
##
##
## ====================================================
## Testing Relationship: mental_issue vs teachersup_sum
## ====================================================
## Predictor Type: NUMERIC -> Running Independent t-test...
##
##
##  Welch Two Sample t-test
##
## data:  d[[col_name]] by d[[target_var]]
## t = 65.45, df = 46520, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##  1.247438 1.324457
## sample estimates:
## mean in group 0 mean in group 1
##        11.71313        10.42718
##
##
## ====================================================
## Testing Relationship: mental_issue vs studsup
## ====================================================
## Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...
##
##
##         0     1
##   0 29098 49000
##   1 16868 12438

##
##  Chi-squared Test for Trend in Proportions
##
## data:  successes out of totals ,
##  using scores: 1 2
## X-squared = 3587, df = 1, p-value < 2.2e-16
##
##
## ====================================================
## Testing Relationship: mental_issue vs studsup_sum
## ====================================================
## Predictor Type: NUMERIC -> Running Independent t-test...
##
##
##  Welch Two Sample t-test
##
## data:  d[[col_name]] by d[[target_var]]
## t = 68.533, df = 45678, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##  1.206009 1.277023
## sample estimates:
## mean in group 0 mean in group 1
##        11.89548        10.65396
```

```
##
##
## ==================================================
## Testing Relationship: mental_issue vs talkf
## ==================================================
## Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...
##
##
##         0     1
##   0 21401 56697
##   1 14959 14347

##
##  Chi-squared Test for Trend in Proportions
##
## data:  successes out of totals ,
##  using scores: 1 2
## X-squared = 5318.8, df = 1, p-value < 2.2e-16
##
##
## ==================================================
## Testing Relationship: mental_issue vs talkm
## ==================================================
## Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...
##
##
##         0     1
##   0  9296 68802
##   1  8470 20836

##
##  Chi-squared Test for Trend in Proportions
##
## data:  successes out of totals ,
##  using scores: 1 2
## X-squared = 4460.4, df = 1, p-value < 2.2e-16
##
##
## ==================================================
## Testing Relationship: mental_issue vs talkscore
## ==================================================
## Predictor Type: NUMERIC -> Running Independent t-test...
##
##
##  Welch Two Sample t-test
##
## data:  d[[col_name]] by d[[target_var]]
## t = -78.768, df = 46097, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##  -1.0182352 -0.9687915
## sample estimates:
## mean in group 0 mean in group 1
##        3.732311        4.725824
```

```
##
##
## ==================================================
## Testing Relationship: mental_issue vs timeexe_r
## ==================================================
## Predictor Type: ORDINAL -> Running Cochran-Armitage Trend Test...
##
##
##          0     1
##    0 19618 58480
##    1 10393 18913

##
##  Chi-squared Test for Trend in Proportions
##
## data:  successes out of totals ,
##  using scores: 1 2
## X-squared = 1132.4, df = 1, p-value < 2.2e-16
##
##
## ==================================================
## Testing Relationship: mental_issue vs timeexe
## ==================================================
## Predictor Type: NUMERIC -> Running Independent t-test...
##
##
##  Welch Two Sample t-test
##
## data:  d[[col_name]] by d[[target_var]]
## t = -33.901, df = 47234, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##  -0.4099063 -0.3650993
## sample estimates:
## mean in group 0 mean in group 1
##        2.900971        3.288473
```

```r
# # chi for nominal
# cont_tbl <- table(d[["mental_issue"]], d[["pmsu_lmh"]])
# print(cont_tbl)
# chi <- chisq.test(cont_tbl)
# print(chi)
# print(chi$p.value)
# print(chi$statistic)
#
# # cochran for ordered
# cont_tbl <- table(d[["mental_issue"]], d[["pmsu_lmh"]])
# cont_tbl
# successes <- cont_tbl["1", ]
# successes
# totals <- colSums(cont_tbl)
# coch <- prop.trend.test(successes, totals)
# coch
#
```

```r
# # t.test for num
# tt <- t.test(d[["pmsu_sum"]] ~ d[["mental_issue"]])
# tt
```

```r
# --- 1. SETUP YOUR TARGET VARIABLE NAME ---
# Target variable is set to "mental_issue" and dataset is named "d"
target_var <- "mental_issue"

# Ensure target is a factor and identify the "success" level (the 2nd level)
d[[target_var]] <- as.factor(d[[target_var]])
target_levels <- levels(d[[target_var]])
success_level <- target_levels[2] # Grabs group 1 / "Yes" status automatically

# --- 2. INITIALIZE AN EMPTY DATAFRAME FOR THE SUMMARY TABLE ---
summary_table <- data.frame(
  Predictor_Variable = character(),
  Variable_Type      = character(),
  Statistical_Test   = character(),
  Test_Statistic     = character(),
  p_value            = character(),
  stringsAsFactors   = FALSE
)

# --- 3. THE AUTOMATED LOOP ---
for (col_name in names(d)) {

  # Skip the target variable itself
  if (col_name == target_var) next

  # Create baseline variables for storing this row's data
  var_type <- "Unknown"
  test_name <- "None"
  stat_val <- "N/A"
  p_val_str <- "N/A"

  # --- 1. IF ORDINAL -> RUN COCHRAN-ARMITAGE TREND TEST ---
  if (is.ordered(d[[col_name]])) {
    var_type <- "Ordinal"
    test_name <- "Cochran-Armitage Trend Test"

    contingency_table <- table(d[[target_var]], d[[col_name]])
    successes <- contingency_table[success_level, ]
    totals    <- colSums(contingency_table)

    test_result <- tryCatch({ prop.trend.test(successes, totals) }, error = function(e) { return(NULL) }

    if (!is.null(test_result)) {
      stat_val <- paste0("X-squared = ", round(test_result$statistic, 2))
      p_val_str <- if(test_result$p.value < 0.001) "< 0.001" else format(round(test_result$p.value, 4),
    }

  # --- 2. IF NUMERIC -> RUN T-TEST ---
  } else if (is.numeric(d[[col_name]])) {
```

```r
    var_type <- "Numeric"
    test_name <- "Welch Two Sample t-test"

    test_result <- tryCatch({ t.test(d[[col_name]] ~ d[[target_var]]) }, error = function(e) { return(NU

    if (!is.null(test_result)) {
      stat_val <- paste0("t(", round(test_result$parameter, 1), ") = ", round(test_result$statistic, 2))
      p_val_str <- if(test_result$p.value < 0.001) "< 0.001" else format(round(test_result$p.value, 4),
    }

  # --- 3. IF NOMINAL / BINARY -> RUN CHI-SQUARE ---
  } else if (is.factor(d[[col_name]]) || is.character(d[[col_name]])) {
    var_type <- if(length(unique(d[[col_name]])) == 2) "Binary" else "Nominal"
    test_name <- "Pearson's Chi-Square Test"

    contingency_table <- table(d[[target_var]], d[[col_name]])
    test_result <- tryCatch({ chisq.test(contingency_table) }, error = function(e) { return(NULL) })

    if (!is.null(test_result)) {
      stat_val <- paste0("X-squared(", test_result$parameter, ") = ", round(test_result$statistic, 2))
      p_val_str <- if(test_result$p.value < 0.001) "< 0.001" else format(round(test_result$p.value, 4),
    }
  }

  # --- 4. APPEND THE RESULT ROW TO OUR TABLE ---
  new_row <- data.frame(
    Predictor_Variable = col_name,
    Variable_Type      = var_type,
    Statistical_Test   = test_name,
    Test_Statistic     = stat_val,
    p_value            = p_val_str,
    stringsAsFactors   = FALSE
  )
  summary_table <- rbind(summary_table, new_row)
}

# --- 4. VIEW THE FINAL SUMMARY TABLE ---
print(summary_table)
```

```
##    Predictor_Variable Variable_Type            Statistical_Test
## 1           seqno_int       Numeric     Welch Two Sample t-test
## 2        country_name       Nominal   Pearson's Chi-Square Test
## 3          bodyheight       Numeric     Welch Two Sample t-test
## 4          bodyweight       Numeric     Welch Two Sample t-test
## 5                MBMI       Numeric     Welch Two Sample t-test
## 6               IOTF4       Ordinal Cochran-Armitage Trend Test
## 7              agecat       Ordinal Cochran-Armitage Trend Test
## 8                 sex        Binary   Pearson's Chi-Square Test
## 9        IRRELFAS_LMH       Ordinal Cochran-Armitage Trend Test
## 10              IRFAS       Numeric     Welch Two Sample t-test
## 11        lifesat_low       Ordinal Cochran-Armitage Trend Test
## 12         mental_sum       Numeric     Welch Two Sample t-test
## 13           pmsu_lmh       Ordinal Cochran-Armitage Trend Test
```

```
## 14          pmsu_sum       Numeric     Welch Two Sample t-test
## 15          cbullied       Ordinal Cochran-Armitage Trend Test
## 16        bulliedsum       Numeric     Welch Two Sample t-test
## 17        onlfreq_is       Ordinal Cochran-Armitage Trend Test
## 18       onlfreq_sum       Numeric     Welch Two Sample t-test
## 19           onlpref       Ordinal Cochran-Armitage Trend Test
## 20       onlpref_sum       Numeric     Welch Two Sample t-test
## 21        famsup_hml       Ordinal Cochran-Armitage Trend Test
## 22        famsup_sum       Numeric     Welch Two Sample t-test
## 23        frisup_hml       Ordinal Cochran-Armitage Trend Test
## 24        frisup_sum       Numeric     Welch Two Sample t-test
## 25        teachersup       Ordinal Cochran-Armitage Trend Test
## 26    teachersup_sum       Numeric     Welch Two Sample t-test
## 27           studsup       Ordinal Cochran-Armitage Trend Test
## 28       studsup_sum       Numeric     Welch Two Sample t-test
## 29             talkf       Ordinal Cochran-Armitage Trend Test
## 30             talkm       Ordinal Cochran-Armitage Trend Test
## 31         talkscore       Numeric     Welch Two Sample t-test
## 32         timeexe_r       Ordinal Cochran-Armitage Trend Test
## 33           timeexe       Numeric     Welch Two Sample t-test
##             Test_Statistic p_value
## 1          t(51650) = -12.6 < 0.001
## 2  X-squared(36) = 2776.08 < 0.001
## 3        t(57278.4) = -7.54 < 0.001
## 4       t(53178.4) = -20.02 < 0.001
## 5       t(49183.8) = -24.23 < 0.001
## 6        X-squared = 105.21 < 0.001
## 7        X-squared = 1151.5 < 0.001
## 8   X-squared(1) = 2430.11 < 0.001
## 9         X-squared = 85.88 < 0.001
## 10       t(52225.8) = 19.65 < 0.001
## 11      X-squared = 8997.69 < 0.001
## 12    t(36694.3) = -518.82 < 0.001
## 13      X-squared = 5875.59 < 0.001
## 14      t(44860.2) = -75.82 < 0.001
## 15      X-squared = 2961.27 < 0.001
## 16      t(38593.8) = -57.63 < 0.001
## 17      X-squared = 1018.32 < 0.001
## 18      t(51786.7) = -21.65 < 0.001
## 19      X-squared = 1621.46 < 0.001
## 20      t(48582.5) = -40.24 < 0.001
## 21      X-squared = 2998.45 < 0.001
## 22       t(47945.3) = 61.31 < 0.001
## 23      X-squared = 1241.26 < 0.001
## 24          t(49170) = 35.4 < 0.001
## 25      X-squared = 3121.83 < 0.001
## 26       t(46519.5) = 65.45 < 0.001
## 27      X-squared = 3586.98 < 0.001
## 28       t(45678.3) = 68.53 < 0.001
## 29      X-squared = 5318.78 < 0.001
## 30      X-squared = 4460.45 < 0.001
## 31        t(46097) = -78.77 < 0.001
## 32      X-squared = 1132.43 < 0.001
## 33       t(47234.5) = -33.9 < 0.001
```

```r
# # create modeling dat
# dat <- dat %>%
#   mutate(y = ifelse(lifesat >5,1,0),
#          y = factor(y))

d_mdl <- d %>%
  select(country_name, MBMI, agecat, sex, IRRELFAS_LMH, mental_issue, pmsu_lmh, cbullied, onlfreq_is, o

#baseline
d_mdl %>%
  group_by(mental_issue) %>%
  tally() %>%
  mutate(prop = n/sum(n))
```

```
## # A tibble: 2 x 3
##   mental_issue     n  prop
##   <ord>        <int> <dbl>
## 1 0            78098 0.727
## 2 1            29306 0.273
```

```r
# data split
set.seed(67)
data_split <- initial_split(d_mdl, prop = 0.8, strata = mental_issue)
train_data <- training(data_split)
test_data  <- testing(data_split)

# 3. Create a Preprocessing Recipe -----------------------------------------
# Define formula (Outcome ~ Predictors) and add preprocessing steps
lr_recipe <- recipe(mental_issue ~ . ,
                    data = train_data) %>%
  step_rm(country_name)
 # step_zv(all_predictors()) %>%          # Remove zero-variance predictors
#  step_normalize(all_numeric_predictors()) # Center and scale numeric variables

# 4. Define the Logistic Regression Model Spec ------------------------------
lr_spec <- logistic_reg() %>%
  set_engine("glm") %>%                   # Use standard generalized linear model
  set_mode("classification")              # Specify binary classification mode

# 5. Create the Workflow ---------------------------------------------------
# Bundle the recipe and model specification together
lr_workflow <- workflow() %>%
  add_recipe(lr_recipe) %>%
  add_model(lr_spec)

# 6. Fit the Model (Train) -------------------------------------------------
# This automatically preps the recipe and trains the model on train data
lr_fit <- fit(lr_workflow, data = train_data)

# 7. Evaluate on Test Data -------------------------------------------------
# Predict classes (labels)
predictions_class <- predict(lr_fit, new_data = test_data)
predictions_class
```

```
## # A tibble: 21,482 x 1
##    .pred_class
##    <ord>
##  1 0
##  2 0
##  3 0
##  4 0
##  5 0
##  6 0
##  7 0
##  8 0
##  9 0
## 10 0
## # i 21,472 more rows
```

```r
# Predict probabilities
predictions_prob  <- predict(lr_fit, new_data = test_data, type = "prob")
predictions_prob
```

```
## # A tibble: 21,482 x 2
##    .pred_0 .pred_1
##      <dbl>   <dbl>
##  1   0.768  0.232
##  2   0.826  0.174
##  3   0.902  0.0976
##  4   0.765  0.235
##  5   0.943  0.0566
##  6   0.873  0.127
##  7   0.884  0.116
##  8   0.846  0.154
##  9   0.902  0.0984
## 10   0.834  0.166
## # i 21,472 more rows
```

```r
# Combine test data with predictions
evaluation_data <- test_data %>%
  bind_cols(predictions_class, predictions_prob)

# Print confusion matrix and accuracy metrics
evaluation_data %>% conf_mat(truth = mental_issue, estimate = .pred_class)
```

```
##           Truth
## Prediction     0     1
##          0 14615  4092
##          1  1005  1770
```

```r
evaluation_data %>% metrics(truth = mental_issue, estimate = .pred_class)
```

```
## # A tibble: 2 x 3
##   .metric  .estimator .estimate
##   <chr>    <chr>          <dbl>
## 1 accuracy binary         0.763
## 2 kap      binary         0.284
```

```r
# Extract and print clean model coefficients
lr_fit %>%
  extract_fit_parsnip() %>%
  tidy() %>%
  print(n = Inf)
```

```
## # A tibble: 22 x 5
##    term            estimate std.error statistic   p.value
##    <chr>              <dbl>     <dbl>     <dbl>     <dbl>
##  1 (Intercept)     -0.708     0.0528    -13.4   4.53e- 41
##  2 MBMI             0.0180    0.00242     7.41  1.24e- 13
##  3 agecat.L         0.0937    0.0166      5.66  1.51e-  8
##  4 agecat.Q         0.0372    0.0146      2.56  1.05e-  2
##  5 sex2             0.550     0.0175     31.4   3.20e-216
##  6 IRRELFAS_LMH.L   0.00201   0.0196      0.103 9.18e-  1
##  7 IRRELFAS_LMH.Q   0.00257   0.0141      0.182 8.56e-  1
##  8 pmsu_lmh.L       0.652     0.0226     28.9   1.20e-183
##  9 pmsu_lmh.Q      -0.0701    0.0164     -4.28  1.87e-  5
## 10 cbullied.L       0.400     0.0168     23.9   8.01e-126
## 11 onlfreq_is.L     0.201     0.0124     16.2   1.00e- 58
## 12 onlpref.L        0.170     0.0161     10.6   4.00e- 26
## 13 onlpref.Q        0.117     0.0154      7.59  3.20e- 14
## 14 famsup_hml.L    -0.222     0.0209    -10.6   2.36e- 26
## 15 famsup_hml.Q    -0.206     0.0215     -9.56  1.21e- 21
## 16 frisup_hml.L    -0.214     0.0196    -10.9   9.49e- 28
## 17 frisup_hml.Q    -0.0487    0.0186     -2.61  9.00e-  3
## 18 teachersup.L    -0.234     0.0127    -18.4   1.33e- 75
## 19 studsup.L       -0.299     0.0126    -23.7   1.16e-124
## 20 talkf.L         -0.287     0.0132    -21.8   4.59e-105
## 21 talkm.L         -0.311     0.0160    -19.4   1.02e- 83
## 22 timeexe_r.L     -0.155     0.0130    -12.0   5.99e- 33
```

```r
# Print coefficients as Odds Ratios
lr_fit %>%
  extract_fit_parsnip() %>%
  tidy(exponentiate = TRUE) %>%
  mutate(is_sig = ifelse(p.value < .05, 1, 0)) %>%
  print(n = Inf)
```

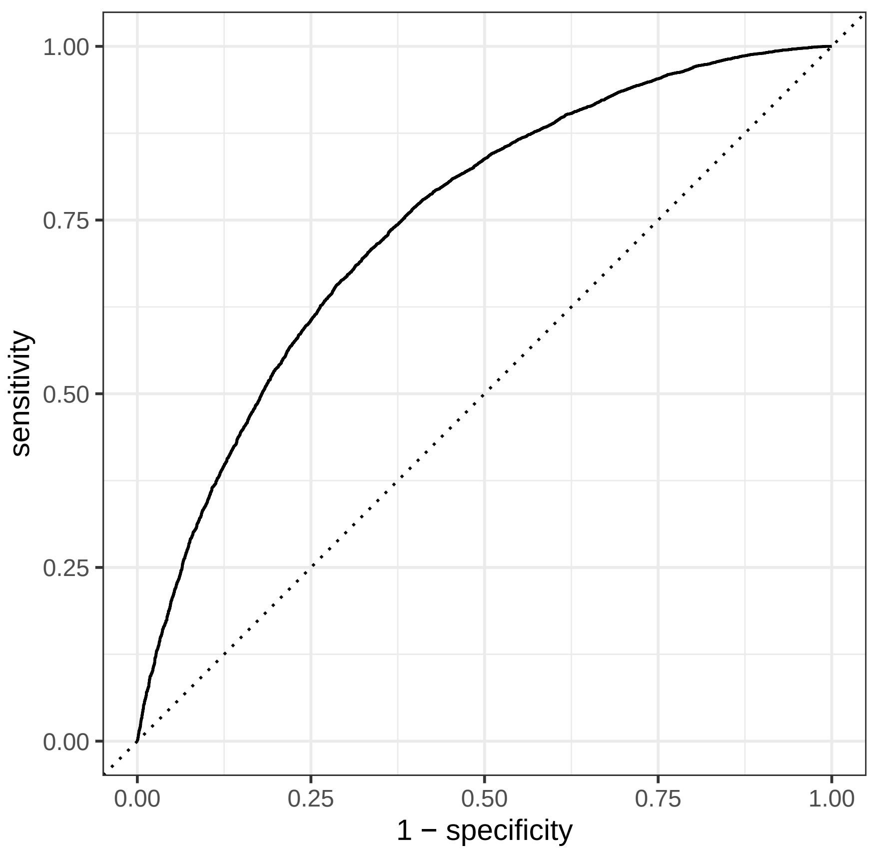
```
## # A tibble: 22 x 6
##    term            estimate std.error statistic   p.value is_sig
##    <chr>              <dbl>     <dbl>     <dbl>     <dbl>  <dbl>
##  1 (Intercept)        0.493   0.0528    -13.4   4.53e- 41      1
##  2 MBMI               1.02    0.00242     7.41  1.24e- 13      1
##  3 agecat.L           1.10    0.0166      5.66  1.51e-  8      1
##  4 agecat.Q           1.04    0.0146      2.56  1.05e-  2      1
##  5 sex2               1.73    0.0175     31.4   3.20e-216      1
##  6 IRRELFAS_LMH.L     1.00    0.0196      0.103 9.18e-  1      0
##  7 IRRELFAS_LMH.Q     1.00    0.0141      0.182 8.56e-  1      0
##  8 pmsu_lmh.L         1.92    0.0226     28.9   1.20e-183      1
##  9 pmsu_lmh.Q         0.932   0.0164     -4.28  1.87e-  5      1
## 10 cbullied.L         1.49    0.0168     23.9   8.01e-126      1
## 11 onlfreq_is.L       1.22    0.0124     16.2   1.00e- 58      1
```

```
## 12 onlpref.L         1.19     0.0161      10.6   4.00e- 26       1
## 13 onlpref.Q         1.12     0.0154       7.59  3.20e- 14       1
## 14 famsup_hml.L      0.801    0.0209     -10.6   2.36e- 26       1
## 15 famsup_hml.Q      0.814    0.0215      -9.56  1.21e- 21       1
## 16 frisup_hml.L      0.807    0.0196     -10.9   9.49e- 28       1
## 17 frisup_hml.Q      0.952    0.0186      -2.61  9.00e-  3       1
## 18 teachersup.L      0.791    0.0127     -18.4   1.33e- 75       1
## 19 studsup.L         0.742    0.0126     -23.7   1.16e-124       1
## 20 talkf.L           0.751    0.0132     -21.8   4.59e-105       1
## 21 talkm.L           0.733    0.0160     -19.4   1.02e- 83       1
## 22 timeexe_r.L       0.856    0.0130     -12.0   5.99e- 33       1
```

```r
# Calculate ROC AUC
evaluation_data %>%
  roc_auc(truth = mental_issue, .pred_0)
```

```
## # A tibble: 1 x 3
##   .metric .estimator .estimate
##   <chr>   <chr>          <dbl>
## 1 roc_auc binary         0.748
```

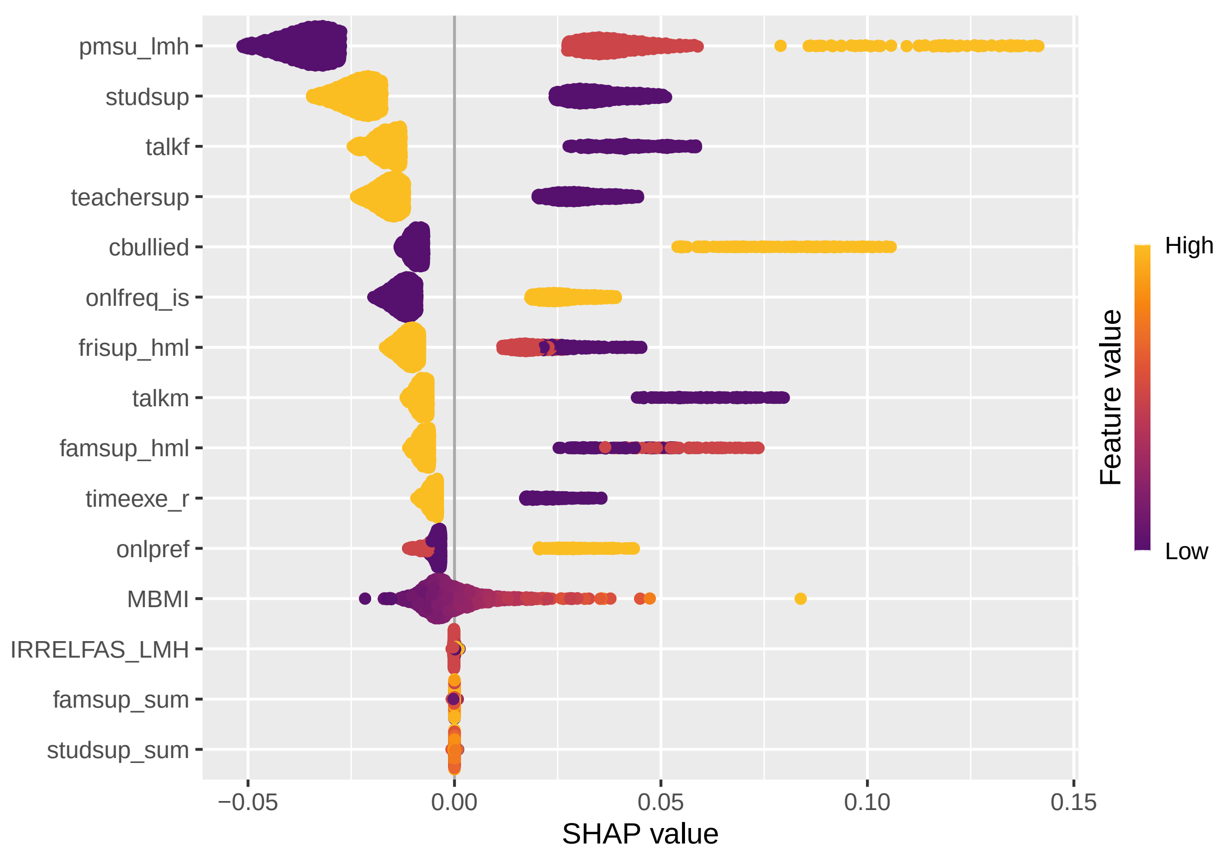
```r
# Generate and plot the ROC Curve
evaluation_data %>%
  roc_curve(truth = mental_issue, .pred_0) %>%
  autoplot()
```



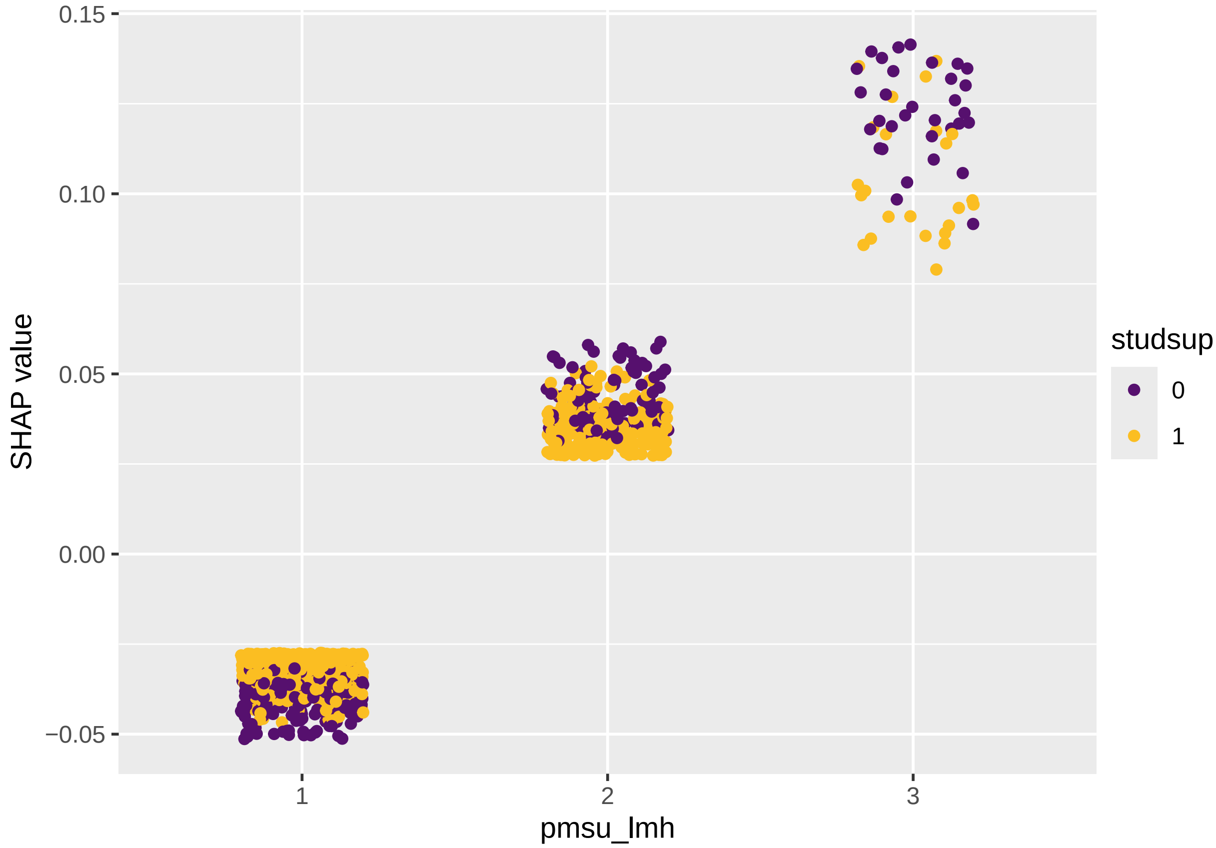
```r
# 1. Isolate just your predictor columns (exclude your target "mental_issue")
# Hint: Use a subset of up to 1,000-2,000 rows for speed
X_explain <- d[1:1000, ] %>% select(-mental_issue)

require(kernelshap)
require(shapviz)

# 2. Compute the SHAP values
# Type = "prob" ensures R looks at the predicted probabilities (e.g., .pred_Yes)
shap_results <- kernelshap(
  object = lr_fit,
  X = X_explain,
  bg_X = d[1:200, ] %>% select(-mental_issue), # Small background baseline
  type = "prob"
)
```

```
##   |                                                                          |
```

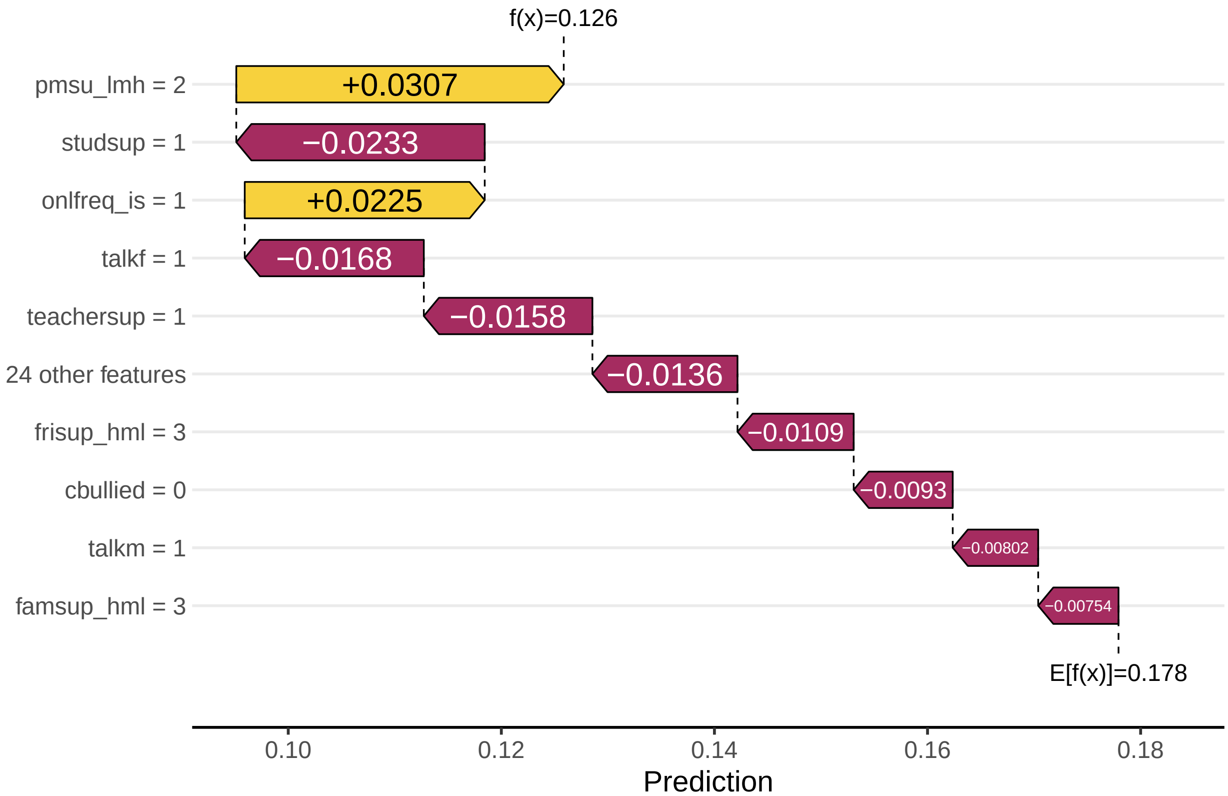
```r
# 3. Convert to a shapviz object for instant plotting
shp_viz <- shapviz(shap_results)

# Replace ".pred_1" with your positive class prediction column name if different
sv_importance(shp_viz$.pred_1, kind = "beeswarm")
```

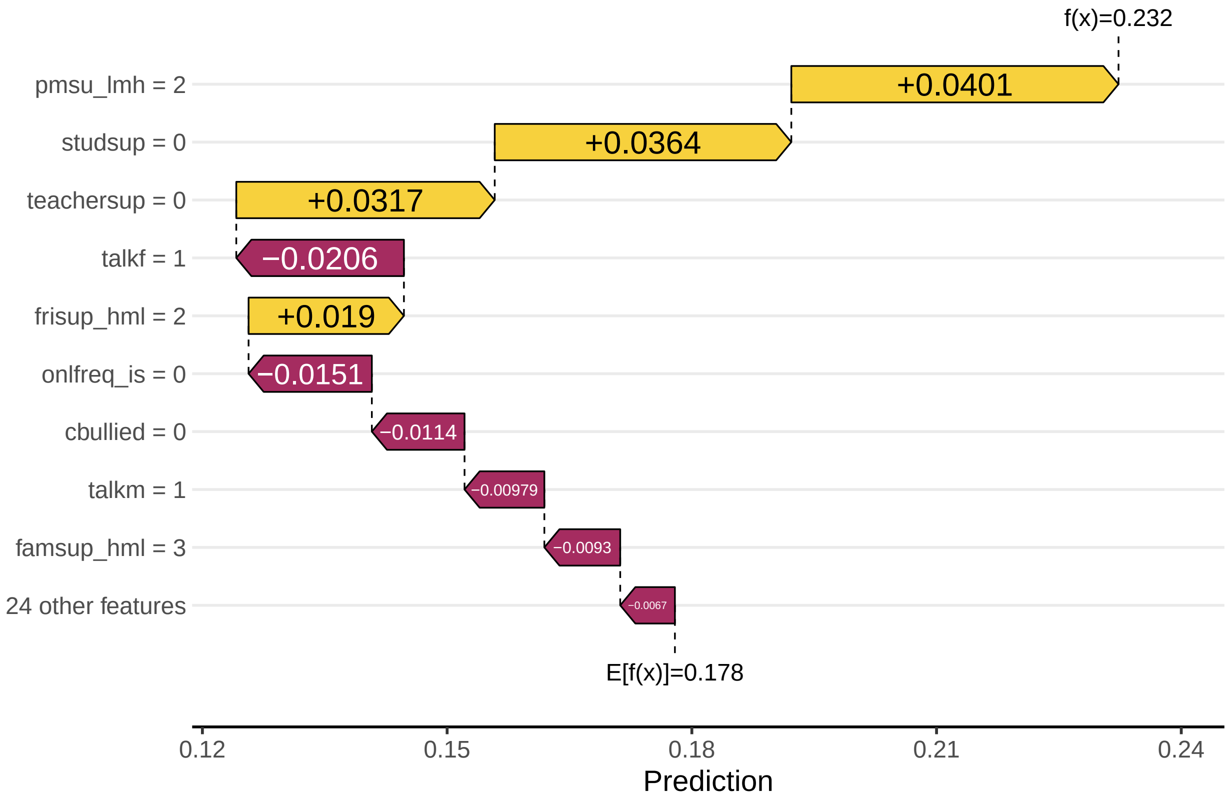


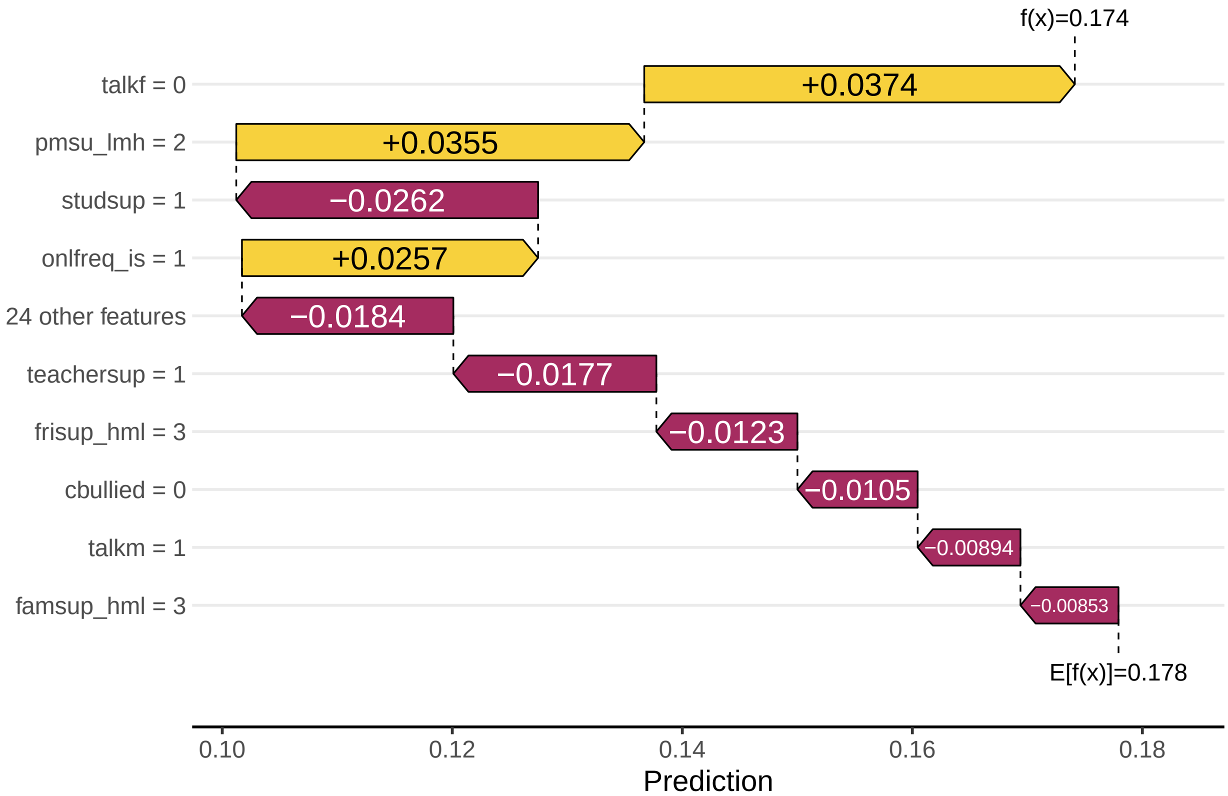
```r
sv_dependence(shp_viz$.pred_1, v = "pmsu_lmh")
```



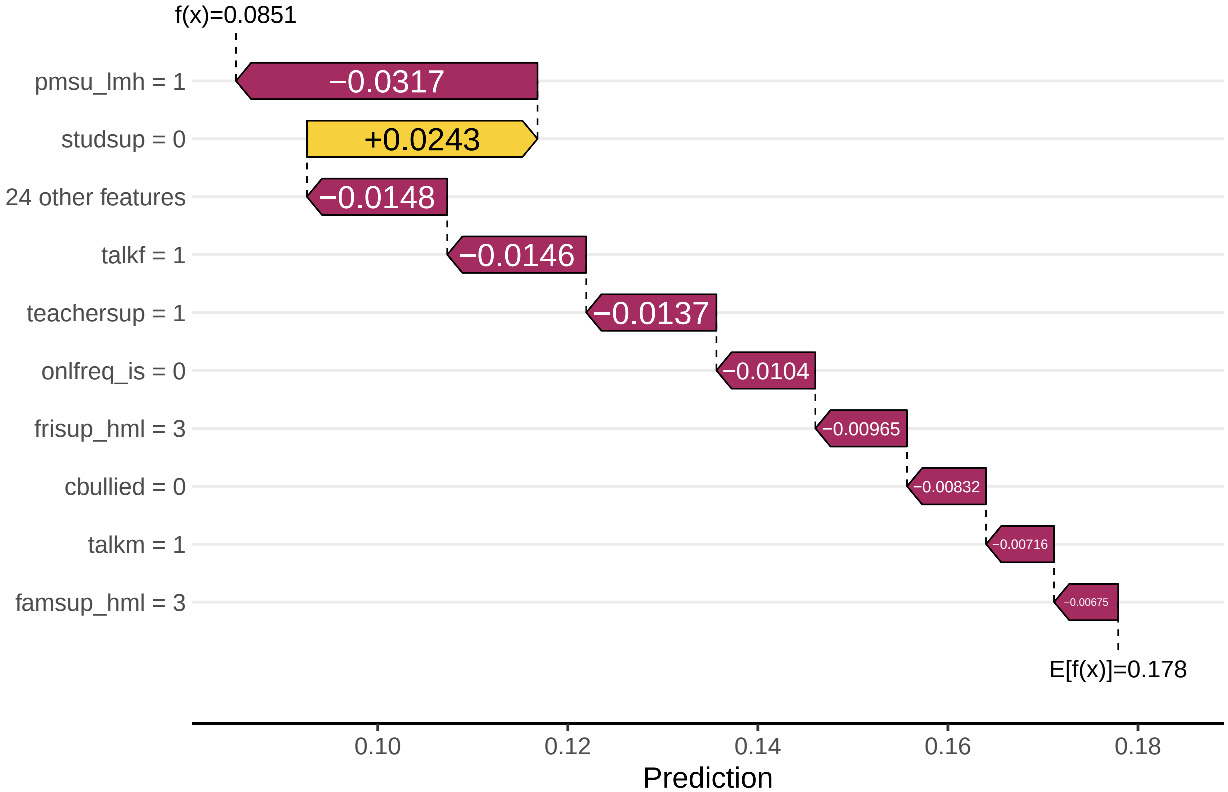

```r
for (i in 1:10){
print(sv_waterfall(shp_viz$.pred_1, row_id = i))
}
```


f(x)=0.126
pmsu_lmh = 2
+0.0307
studsup = 1
−0.0233
onlfreq_is = 1
+0.0225
talkf = 1
−0.0168
teachersup = 1
−0.0158
24 other features
−0.0136
frisup_hml = 3
−0.0109
cbullied = 0
−0.0093
talkm = 1
−0.00802
famsup_hml = 3
−0.00754
E[f(x)]=0.178
0.10
0.12
0.14
0.16
0.18
Prediction

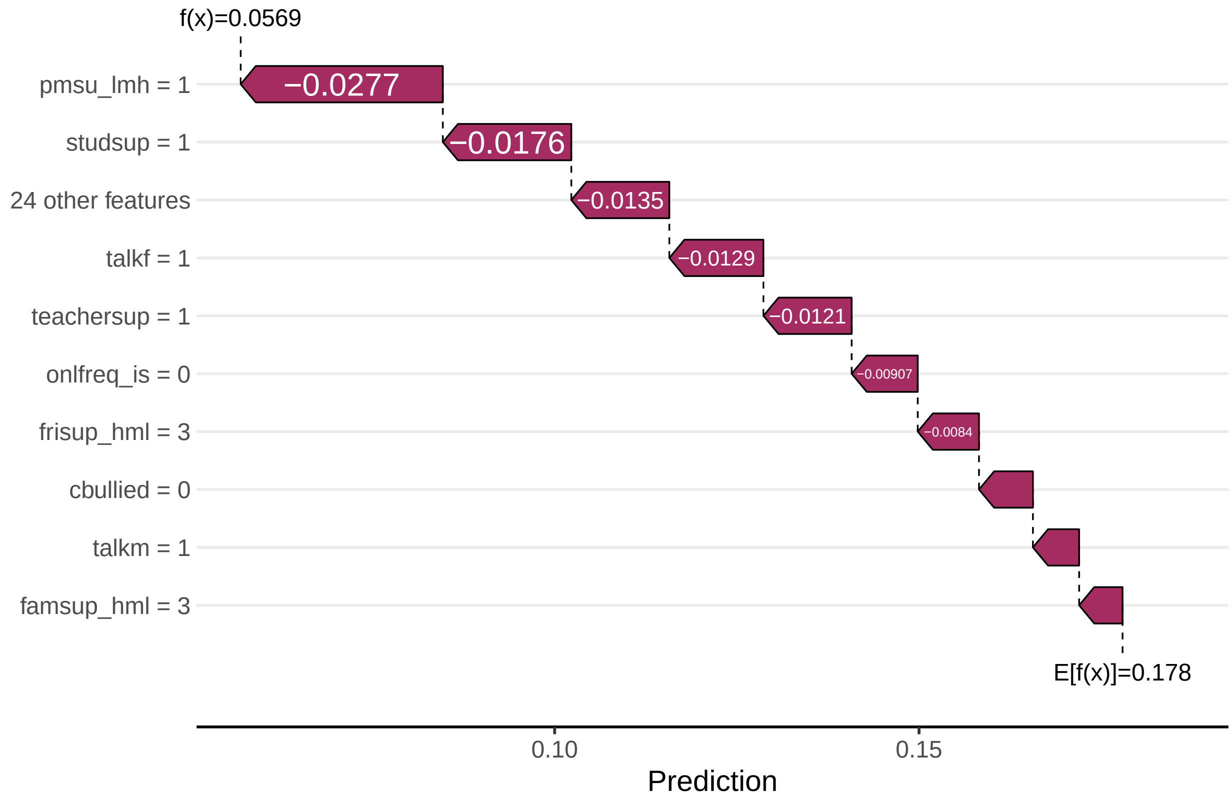

f(x)=0.232
pmsu_lmh = 2
+0.0401
studsup = 0
+0.0364
teachersup = 0
+0.0317
talkf = 1
−0.0206
frisup_hml = 2
+0.019
onlfreq_is = 0
−0.0151
cbullied = 0
−0.0114
talkm = 1
−0.00979
famsup_hml = 3
−0.0093
24 other features
−0.0067
E[f(x)]=0.178
0.12
0.15
0.18
0.21
0.24
Prediction

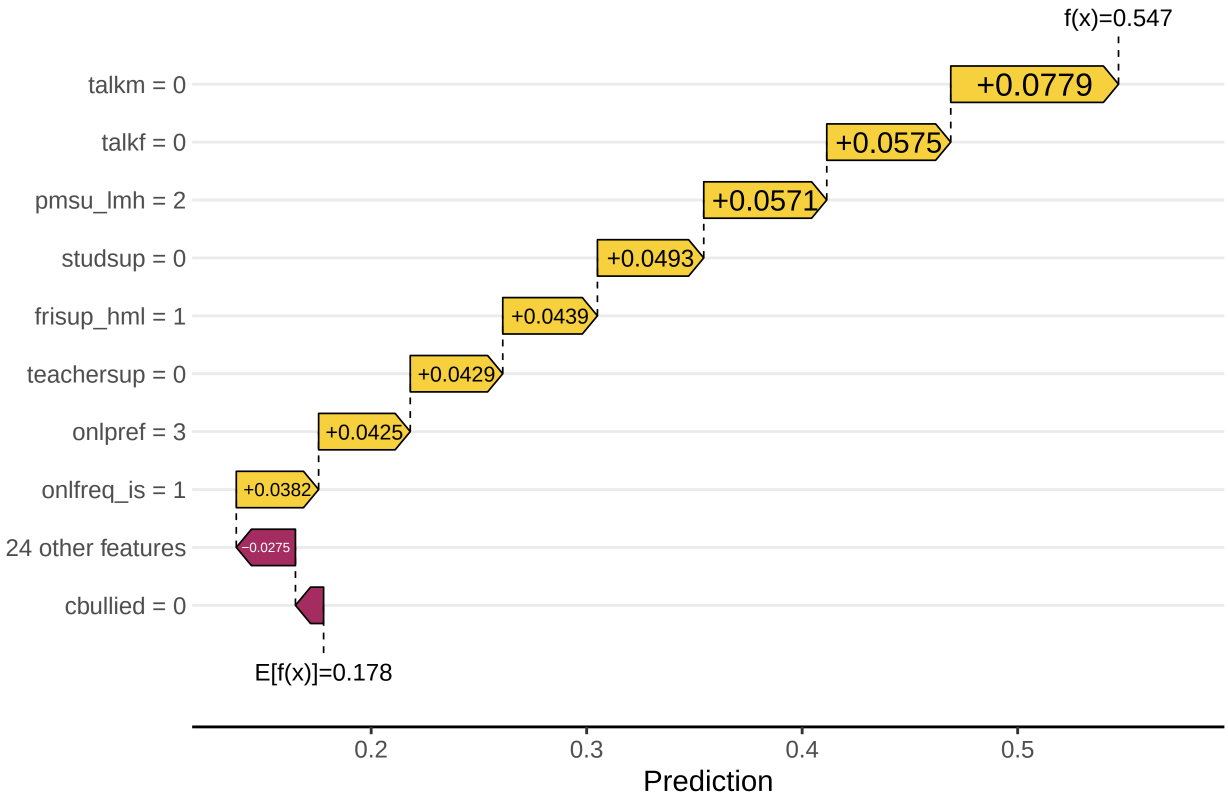


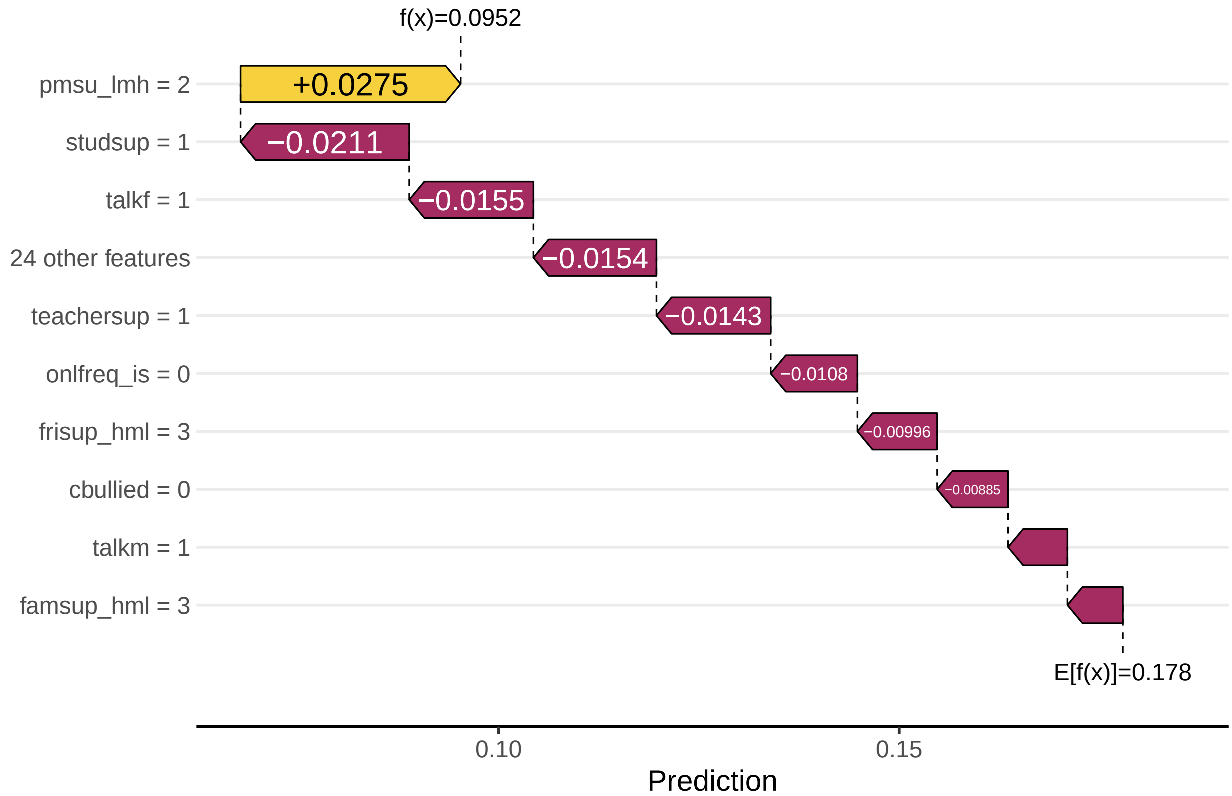
f(x)=0.174
talkf = 0
+0.0374
pmsu_lmh = 2
+0.0355
studsup = 1
−0.0262
onlfreq_is = 1
+0.0257
24 other features
−0.0184
teachersup = 1
−0.0177
frisup_hml = 3
−0.0123
cbullied = 0
−0.0105
talkm = 1
−0.00894
famsup_hml = 3
−0.00853
E[f(x)]=0.178
0.10
0.12
0.14
0.16
0.18
Prediction



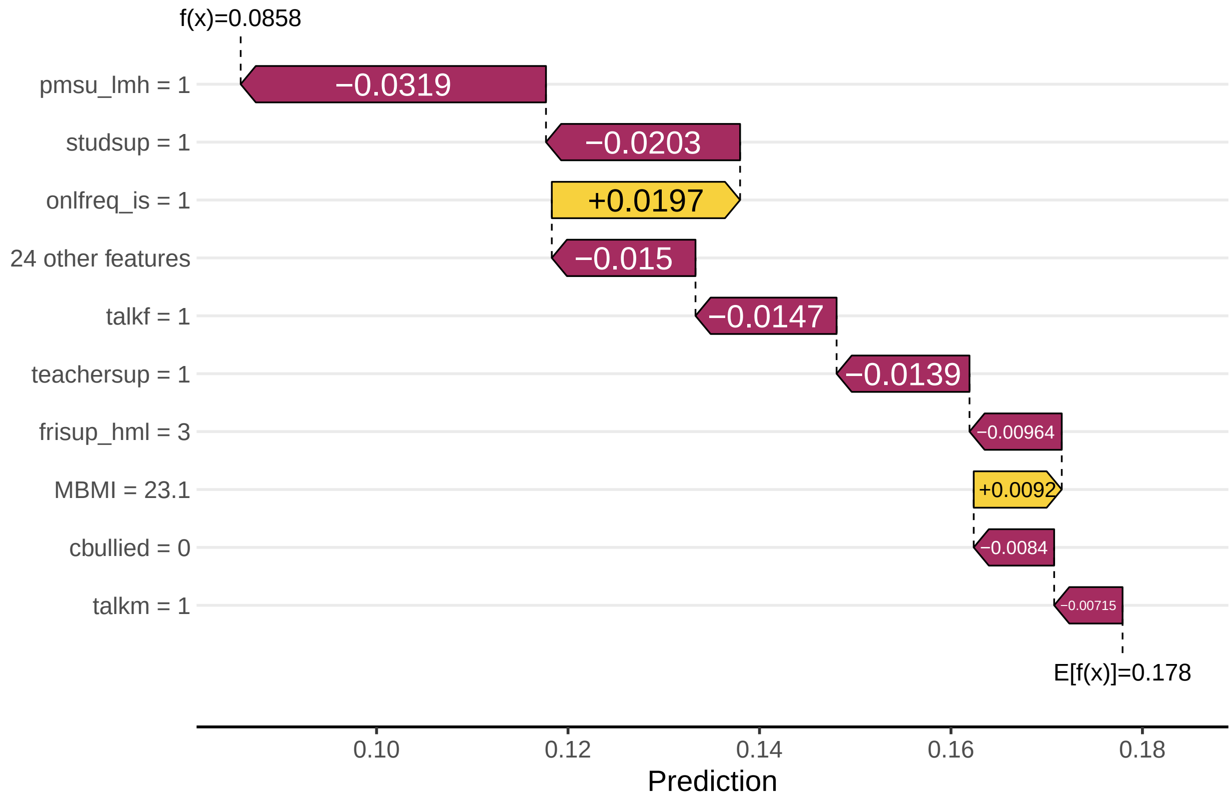
f(x)=0.0851
pmsu_lmh = 1
studsup = 0
24 other features
talkf = 1
teachersup = 1
onlfreq_is = 0
frisup_hml = 3
cbullied = 0
talkm = 1
famsup_hml = 3
−0.0317
+0.0243
−0.0148
−0.0146
−0.0137
−0.0104
−0.00965
−0.00832
−0.00716
−0.00675
E[f(x)]=0.178
0.10
0.12
0.14
0.16
0.18
Prediction

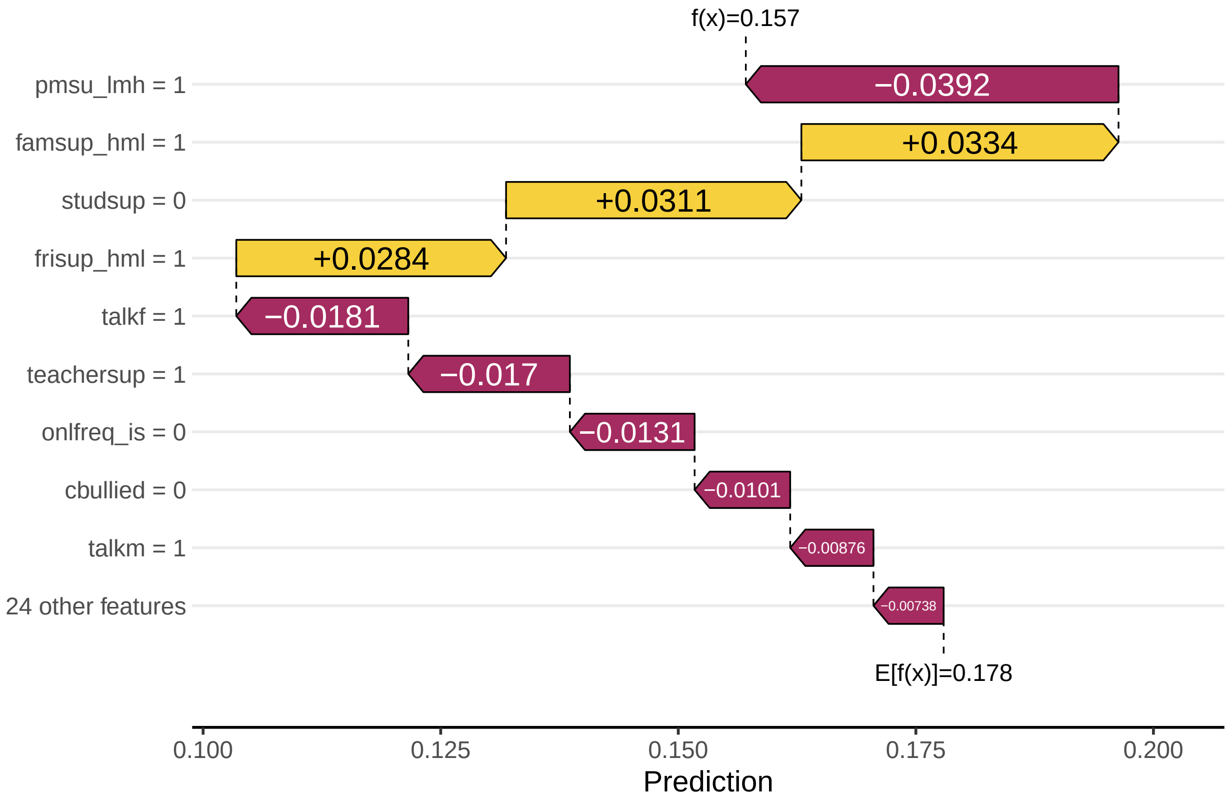

f(x)=0.0569
pmsu_lmh = 1
studsup = 1
24 other features
talkf = 1
teachersup = 1
onlfreq_is = 0
frisup_hml = 3
cbullied = 0
talkm = 1
famsup_hml = 3
−0.0277
−0.0176
−0.0135
−0.0129
−0.0121
−0.00907
−0.0084
E[f(x)]=0.178
0.10
0.15
Prediction

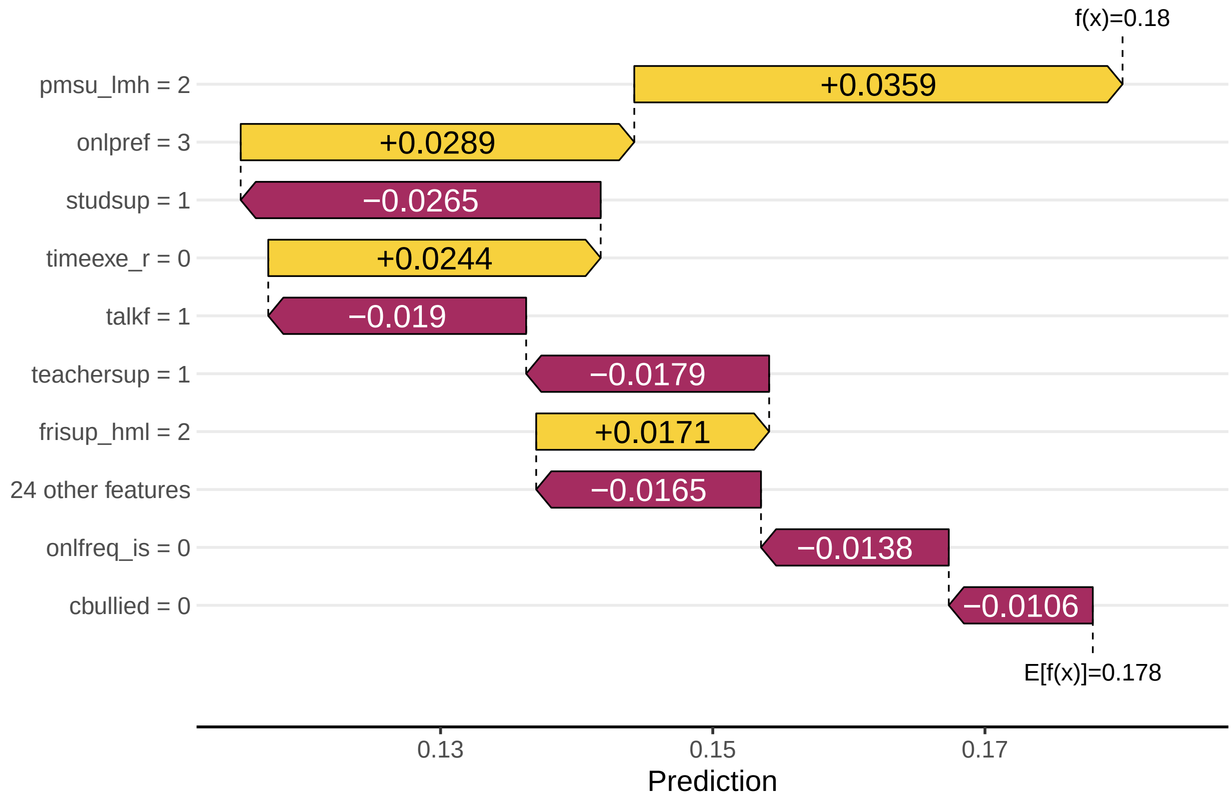

f(x)=0.547
talkm = 0
talkf = 0
pmsu_lmh = 2
studsup = 0
frisup_hml = 1
teachersup = 0
onlpref = 3
onlfreq_is = 1
24 other features
cbullied = 0
+0.0779
+0.0575
+0.0571
+0.0493
+0.0439
+0.0429
+0.0425
+0.0382
−0.0275
E[f(x)]=0.178
0.2
0.3
0.4
0.5
Prediction



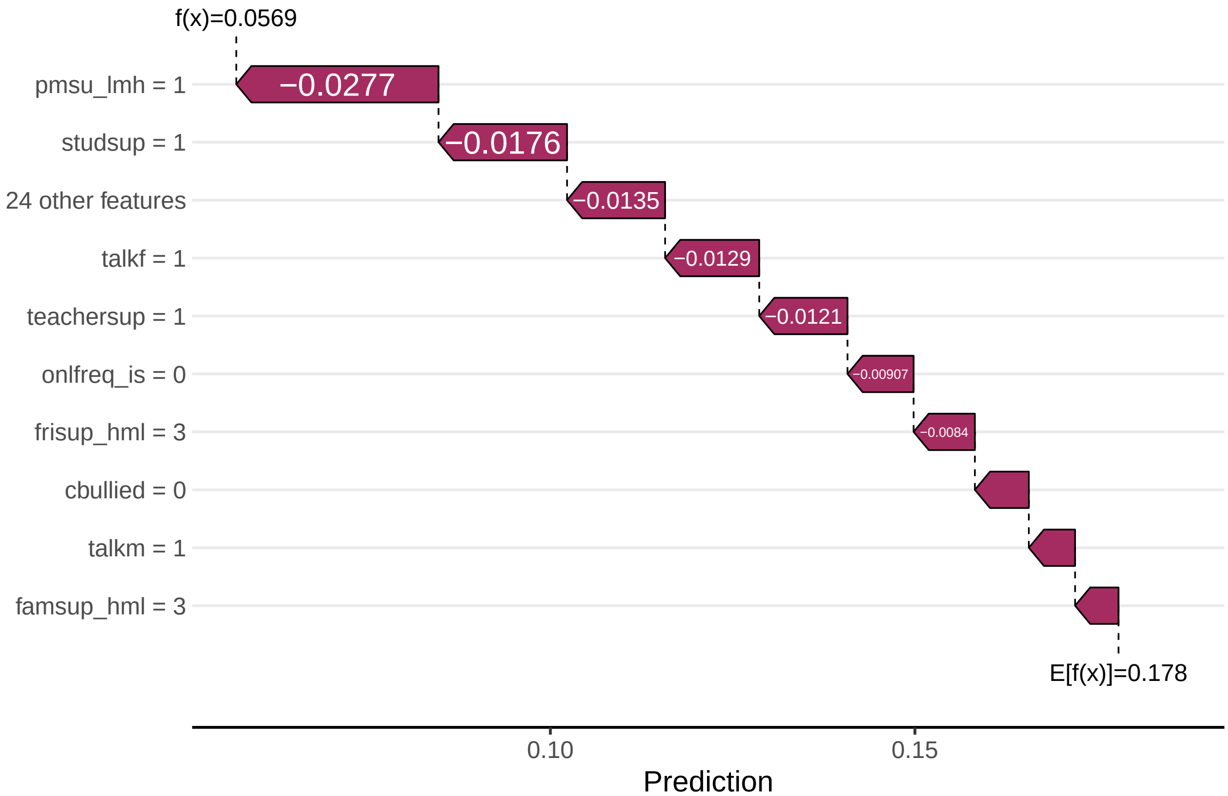
f(x)=0.0952
pmsu_lmh = 2
+0.0275
studsup = 1
−0.0211
talkf = 1
−0.0155
24 other features
−0.0154
teachersup = 1
−0.0143
onlfreq_is = 0
−0.0108
frisup_hml = 3
−0.00996
cbullied = 0
−0.00885
talkm = 1
famsup_hml = 3
E[f(x)]=0.178
0.10
0.15
Prediction


f(x)=0.0858
pmsu_lmh = 1
studsup = 1
onlfreq_is = 1
24 other features
talkf = 1
teachersup = 1
frisup_hml = 3
MBMI = 23.1
cbullied = 0
talkm = 1
−0.0319
−0.0203
+0.0197
−0.015
−0.0147
−0.0139
−0.00964
+0.0092
−0.0084
−0.00715
E[f(x)]=0.178
0.10
0.12
0.14
0.16
0.18
Prediction


f(x)=0.157
pmsu_lmh = 1
famsup_hml = 1
studsup = 0
frisup_hml = 1
talkf = 1
teachersup = 1
onlfreq_is = 0
cbullied = 0
talkm = 1
24 other features
−0.0392
+0.0334
+0.0311
+0.0284
−0.0181
−0.017
−0.0131
−0.0101
−0.00876
−0.00738
E[f(x)]=0.178
0.100
0.125
0.150
0.175
0.200
Prediction




```r
sv_waterfall(shp_viz$.pred_1, row_id = 5)
```


f(x)=0.0569
pmsu_lmh = 1
studsup = 1
24 other features
talkf = 1
teachersup = 1
onlfreq_is = 0
frisup_hml = 3
cbullied = 0
talkm = 1
famsup_hml = 3
−0.0277
−0.0176
−0.0135
−0.0129
−0.0121
−0.00907
−0.0084
E[f(x)]=0.178
0.10
0.15
Prediction